
JOURNAL OF INQUIRY-BASED LEARNING IN MATHEMATICS
No. **57**, (April 2023)

Finite Mathematics

Mindy Capaldi

Valparaiso University

Contents

To the Instructor	iv
1 Linear Equations	1
1.1 Lines	1
1.2 Pairs of Lines	3
1.3 Applications	4
2 Systems of Linear Equations	7
2.1 The Substitution Method	7
2.2 The Elimination Method	9
2.3 The Matrix Method	11
2.4 Systems of m Linear Equations Containing n Variables . . .	15
3 Matrices	18
3.1 Matrix Algebra	18
3.2 Matrix Multiplication	19
3.3 The Inverse of a Matrix	21
4 Linear Programming: Graphical	23
4.1 Systems of Linear Inequalities	23
4.2 A Geometric Approach for Problems with Two Variables . .	24
5 Linear Programming: The Simplex Method	26
5.1 The Simplex Tableau: Pivoting	26
5.2 Maximum Problems in Standard Form	29
5.3 The Duality Principle for Minimum Problems	33
5.4 Non-Standard Form Problems	37
6 Sets and Counting	41
6.1 Sets	41

6.2	The Number of Elements in a Set	44
6.3	The Multiplication Principle	45
6.4	Permutations	46
6.5	Combinations	49
7	Probability	52
7.1	Sample Spaces and Probabilities	52
7.2	Properties of the Probability of an Event	54
7.3	Expected Value	56
7.4	Conditional Probability	57
7.5	Independent Events	59
7.6	Bayes' Theorem	61
8	Markov Chains	63
8.1	Markov Chains and Transition Matrices	63
8.2	Regular Markov Chains	66
8.3	Absorbing Markov Chains	69
9	Two-Person Games	76
9.1	Strictly Determined Games	76
9.2	Mixed Strategies	78
9.3	Finding Optimal Mixed Strategies	79
	Notes to the Instructor	82
	Appendix A	83
	Appendix B	86

To the Instructor

These notes are for a four-credit, 100-level finite mathematics course. The course typically meets for four 50-minute class periods each week. We go through the full set of notes in one semester. For instructors or institutions with less class time, not all chapters or topics may be feasible.

Originally, I developed notes for an online version of the class based on my lecture notes. The online notes were then modified to be inquiry-based for the in-person version of the class. Many of the students in this course are business majors, for whom it is required, but finite mathematics also fulfills the quantitative analysis general education requirement. Students range from freshmen to seniors and have a wide variety of mathematical backgrounds. These notes include a sufficient amount of explanation and detail to serve these students. If an instructor knew that their students had a stronger mathematical foundation, such as a prerequisite statistics course, the notes could be easily updated by leaving out explanatory passages.

The topics covered in this course are matrix algebra, linear programming, probability, counting methods, Markov chains, and game theory. Applications are carefully chosen to be of interest to students and include, but are not limited to, business topics, campus life, politics, and sports. Additionally, to engage students and break up the monotony of the semester, there are two appendices describing board game modules. I was able, through an internal grant, to obtain copies of *Carcassonne* and *Chutes and Ladders: Superhero Edition* for the students to play in class. After a round of play, we go over a problem set related to the game. Students respond positively to these activities, and they could be adapted as course projects or applied to similar board games.

The Simplex Methods chapter is less inquiry-based than the others. I found that for this chapter it was helpful to have guided lectures to introduce some of the material. For example, in problem 102 students were led to the development of a simplex tableau. In small groups, students were given the opportunity to explore and share ideas for how a matrix might be used in the new context. Once possible matrices were written on boards around the room, I built off of their ideas and brought the discussion around to my preferred notation and tableau structure. Individual instructors can decide what level of inquiry is suitable for their class and students in this sort of

manner, for the Simplex Method or other topics.

Student grades are based on participation (15%), homework (20%), three tests (45%), and a final exam (20%). Participation points are accumulated through attendance, presenting at the board, and contributing to class discussion. Board work does not have to be correct to earn points, and I tell the students that I like wrong answers to go up because then we have something to talk about. For this class, I have tried handwritten homework but more recently switched to online homework administered through WebWork. I find that students appreciate the immediate feedback and are more willing to ask questions because there is a simple “email instructor” link for each problem. I allow them to attempt each problem, except some multiple choice exercises, as many times as they wish. The class meets for four 50-minute periods each week. A typical class size is around 30 students.

Prior to the start of the semester, students purchase a spiral-bound copy of the course notes, which also includes the syllabus. I provide a daily schedule, with test dates, homework dates, and drop/withdraw deadlines, as well. Compared to most textbooks, the spiral-bound notes are quite cheap. Students are also more organized and prepared for class, since they mostly just need to bring the note set and paper/pencil. Although I used to motivate the use of inquiry-based learning at the beginning of the semester, I found that students question what is happening in class less if I do not point out that it is unusual. However, after a couple of weeks I offer extra credit for coming to my office and “introducing themselves.” While this is a great chance for me to put names to faces and get to know students better, it is also an opportunity to receive feedback about the class structure. Since they have had time to experience my teaching methods, I get honest and helpful responses. I also have time to adjust the course structure if necessary, because these meetings are relatively early in the semester.

In class, I write announcements on the board and often project the notes on a screen. After going over the announcements, I usually ask a few refresher questions about the previous class period’s lesson. We then pick up in the notes wherever we ended in the last class. I do not leave students to read through the notes on their own and solve problems. Instead, because of the type and level of the students in this course, I read through the text between problems, sometimes elaborate on it, and then give them time to work in groups on problems. By providing some lecture time, however small, students feel supported.

The majority of class time is spent on group work and student presentations. Groups are self selected and usually have 2-5 people. As they attempt problems in their groups I circulate the room, answering or asking questions. If a group is quiet, I might ask them what ideas they have for the problem, or what a certain notation means. As I walk around, I am also able to help students who are struggling more with the material and/or have

nothing written on their paper. When I can see that almost all students have an answer, I have someone go to the board to present their solution. Presentations are usually on a voluntary basis. As a class, we discuss the answer and make any necessary corrections.

I have added comments on particular problems throughout the notes. These superscripts refer to endnotes in the last Chapter titled, “Notes to the Instructor.”

Additional Resources. The daily schedule and/or solutions are available upon request. As long as I can see that you are an instructor at an institution, I am happy to provide the solutions.

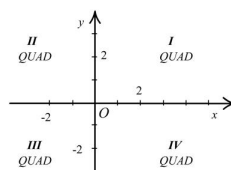
Chapter 1

Linear Equations

1.1 Lines

Rectangular (Cartesian) coordinates, often referred to as the xy -plane, are two axes that are perpendicular (right angles) to each other.

Definition 1. A plane can be divided into four **quadrants** in the following way:



We describe a point on the plane using the notation (x, y) . The first number, the x coordinate, describes how far the point is from the vertical y -axis. The second number, the y coordinate, describes how far the point is from the horizontal x -axis.

Problem 2. What can you tell me about points in quadrant I (positive or negative x and y values)? Quadrant III?

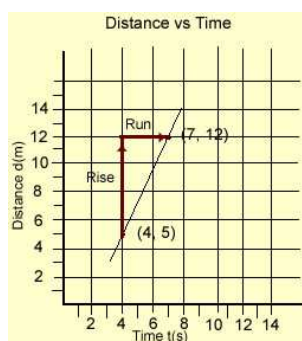
Definition 3. The **slope** of a line containing points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_1 - y_2}{x_1 - x_2}$ for two points, (x_1, y_1) and (x_2, y_2) . This also could be read as $\frac{\text{change in } y}{\text{change in } x}$.

There are four ways slope can be identified:

- Negative slope - the line moves downward from left to right as x increases (hint: think of the diagonal in “N”)

- Positive slope - the line moves upward from left to right as x increases
- Zero slope - horizontal line
- Undefined slope (denominator 0)- vertical line

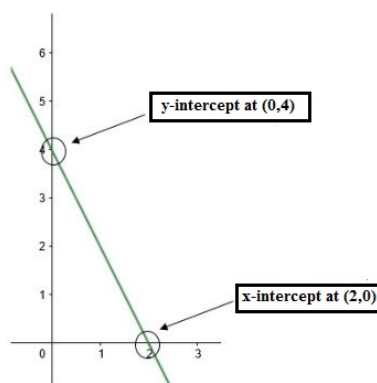
Problem 4. Find the slope of the line below. Describe what the slope represents in terms of time and distance.



Problem 5. Would using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ give a different slope? Why or why not? Does it matter which point is chosen to be (x_1, y_1) and which is chosen to be (x_2, y_2) ?

There are two main ways that we will write the equation of a line in this class. To understand the first, you need to know what an intercept is. Think about the word “intercept”. If you intercepted a football, then you caught the ball.

Definition 6. A *y-intercept* is the point where the line crosses (or catches) the y -axis. Similarly, an *x-intercept* is the point where the line crosses the x -axis.



Definition 7. A linear equation is an equation for a straight line. Two forms of Linear Equations are:

- **Slope-Intercept Form:** $y = mx + b$, where $m = \text{slope}$ and $b = \text{y-intercept}$
- **Point-Slope Form:** $y - y_1 = m(x - x_1)$, where $m = \text{slope}$ and (x_1, y_1) is a point on the line

Problem 8. Given the linear equation $y = mx + b$, where m is not 0 or undefined, what must x equal for the line to intersect, or cross, the y -axis? What must y equal for the line to intersect the x -axis?

Problem 9. Find the x and y intercepts of the following lines, if they exist:

a) $2x + 4y = 8$

b) $x = 3$

c) $y = 0$

Problem 10. Find the slope of each line, or determine that it is undefined, and then graph the line:

a) $y = -\frac{3}{2}x + 3$

b) The line through the points $(-3, 2)$ and $(0, 4)$.

c) $x = 4$

d) $y + 3 = \frac{1}{2}(x - 6)$

1.2 Pairs of Lines

There are three categories for pairs of lines. They could be intersecting (one point in common), parallel (no points in common), or coincident (all points in common).

Problem 11. For each type, describe how the slopes and y -intercepts of the two lines compare. Are they the same or different? Give an example graph of each type.

Once you have solved the previous problem, write in your answers (same, different, or either) as a definition.

Definition 12. Types of Pairs of Lines:

1. *Intersecting:* Their slopes are _____ and their y -intercepts are _____.
Graph:

2. *Parallel:* Their slopes are _____ and their y-intercepts are _____.

Graph:

3. *Coincident:* Their slopes are _____ and their y-intercepts are _____.

Graph:

Problem 13. The lines $2x - y = 5$ and $x + y = 4$ are intersecting lines. Solve for y (in terms of x) in each linear equation.

- a) How do you find the point of intersection of the two lines? Go ahead and actually solve for it (without graphing).
- b) Now graph the lines. Does the intersection match what you found for part a?

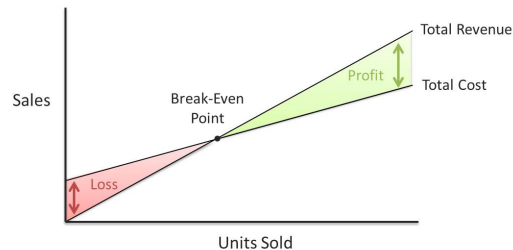
Problem 14. According to *rentcafe.com*, the May 2018 average monthly rent for a 2-bedroom apartment in Chicago, IL was \$2,287. Comparatively, the rent for a similar apartment in Los Angeles, CA was \$2,710.

- a) Using the price of rent, find an equation to express the rent cost, y , in terms of the total months, x , for an apartment in Chicago. Find a similar equation for an apartment in Los Angeles.
- b) Find the point of intersection of each of the lines that you found with the line $y = 25000$. What does it mean for these lines to intersect $y = 25,000$ and what does the difference between those intersections tell us?

1.3 Applications

Consider a business that produces x units of a product. It costs C dollars to make the units and the business makes a revenue of R dollars from selling the units. The business wants the revenue to exceed the cost ($R > C$). If the revenue is less than the cost ($R < C$), then it is losing money.

Definition 15. The **break-even point** is the point, (x, y) , when revenue and cost are the same ($R = C = y$).



This graph shows two lines, one for cost and one for revenue. They can be written as $R = m_1x$ and $C = m_2x + b$, where m_1 is the revenue slope. The y-intercept for revenue is 0, indicating there is no revenue when no units are sold. Then the slope and y-intercept for cost are m_2 and b , respectively. When no units are sold, the baseline costs of the business are shown by b .

Problem 16. Bob makes hamburgers that cost \$1 to make and sell for \$2. Bob also spends money on salaries, rent, utilities, etc, which run him \$300 a day.

- Find the equation for the cost, C , of making x number of burgers per day.
- Find the equation for the revenue, R , of selling x number of burgers.
- Find the break-even point, (x, y) , that tells us the number of burgers (x) that must be sold for Bob's cost to match his revenue (y).
- Graph C and R and check the answer to part c.

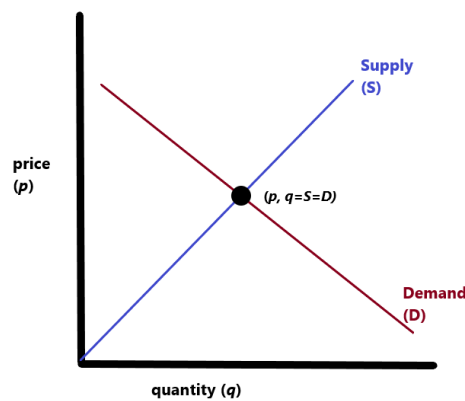
Problem 17. Dana wants to start a summer lawn mowing service. They plan on charging \$15/hour. It costs Dana \$4/hour to run their lawn mower and \$30 per day in other expenses (food, gas for car, etc.). Assume Dana mows for 5 hours most days.

- Determine the cost and revenue equations from Dana's lawn mowing service.
- Find the break-even point and explain what it means to Dana.
- What if Dana charges \$18/hour? Find the new break-even point and state what the difference between this point and the original point tells us.

Instead of cost and revenue, we could similarly solve problems involving supply and demand. The **supply equation**, S , in economics gives the amount of an item that sellers are willing to sell. The **demand equation**,

D , specifies the amount of the item that buyers want to buy. Each of these equations involve the price, p , of the item, and the quantity, q , of the item. As more of the quantity are available, it means demand (D) is decreasing and supply is increasing. You can also think of it this way: if the price goes down, then demand goes up and supply decreases.

Definition 18. Given linear equations for Supply (S) and Demand (D), the **market price** is the value of the price, p , where supply=demand ($S = D$).



Problem 19. If the price of an item was set below the market price, what would that mean for supply compared to demand?

Problem 20. Suppose that the market price for a tablet is \$350 and it occurs when 100,000 units are produced. If the price was \$50, it is not worth it to make the tablet, so no units are made. No one wants to pay \$600 for the tablet, so at that price none would be purchased.

- If we graphed this problem, the information given describes three points on that graph. What are the three points (give the actual x and y coordinates)?
- Find the supply and demand equations, assuming they are linear.
- Graph the supply and demand lines.

Chapter 2

Systems of Linear Equations

2.1 The Substitution Method

In the application section we looked at problems with two linear equations (supply and demand or cost and revenue). Problems such as these, which involve more than one equation, are called systems of equations. Equations with three variables are still not too difficult to graph (think of a third axis coming out of the paper at you), and three variable equations correspond to our 3-Dimensional world. It is harder to visualize or graph equations with more than four variables, although we can use them to describe and model real-world phenomena!

Definition 21. A *system of linear equations* is a collection of two or more linear equations, each containing one or more variables.

Example 22. These are Systems of Linear Equations:

a)

$$\begin{cases} x + 2y = 3 \\ 3x - 3y = 4 \end{cases}$$

b)

$$\begin{cases} -x + 3y - z = 3 \\ x - 4y + 2z = 4 \\ 2x + y = 5 \end{cases}$$

c)

$$\begin{cases} -x_1 + 3x_2 - x_3 = 3 \\ x_1 - 4x_2 + 2x_3 = 4 \\ 2x_1 + x_2 = 5 \end{cases}$$

Notice that the last two systems represent the same equations. The only difference is what the variables have been named (x_1 instead of x , x_2 instead of y , and x_3 instead of z).

Problem 23. *We found previously that there are three types of pairs of lines. If lines are intersecting, how many solutions would there be? In other words, how many points (x,y) could be validly plugged into both equations? What about if they are parallel or coincident?*

Problem 24. *Given*

$$\begin{cases} x - 3y = 4 \\ -2x + 5y = 1 \end{cases}$$

- Solve for x in the first equation.*
- Substitute for x in the second equation using your answer to part a.*
- Now solve for the value of y .*
- Now solve for the value of x .*
- What values of (x,y) give a the solution to the system? Were these lines parallel, coincident, or intersecting?*

That method of solving a system of equations is called the **substitution method** since you solved for one variable in the first equation and then **substituted** it into the second equation. It works well when you only have 2 or 3 equations.

Problem 25. *Given*

$$\begin{cases} y = -x + 3 \\ y = -x + 1 \end{cases}$$

- Without solving this system, how can you tell whether these lines are parallel, coincident, or intersecting?*
- What do you get when solving this system of equations using the substitution method? How does this confirm your answer from part a?*

Problem 26. *Given*

$$\begin{cases} 2y = -2x + 2 \\ y = -x + 1 \end{cases}$$

- Without solving this system, how can you tell whether these lines are parallel, coincident, or intersecting?*
- What do you get when solving this system of equations using the substitution method? How does this confirm your answer from part a?*

Definition 27. *When no solution exists for a system of equations, we say the system is **inconsistent**. If one or more solutions exist, then the system is **consistent**.*

There are three possible types of solutions to a system of equations with two variables, depending on whether the lines are parallel, intersecting, or coincident. Match the type of lines with the appropriate answer below.

Types of Answers:

1. _____ lines are a consistent system with one solution.
2. _____ lines are a consistent system with infinitely many solutions.
3. _____ lines are an inconsistent system with no solutions.

Problem 28. *Make up three systems of two linear equations containing two variables, one that has no solution, one that has exactly one solution, and one that has infinitely many solutions.*¹

2.2 The Elimination Method

Sometimes the substitution method can be tedious. This usually happens if there are more than two variables. In those cases, you may want to use the elimination method. This method also connects to what we will learn in later sections with matrices.

Step 1: Identify a variable that you want to eliminate and two equations that contain that variable.

Example 29. Use the elimination method to solve

$$\begin{cases} x - 2y = 4 \\ 3x + y = 1 \end{cases}$$

Step 1: In step 2, I will want either the x variable or the y variable to have the same coefficient in the two equations. Since neither variable already has the same coefficient in both equations, it does not matter in this case which one is chosen. I choose to eliminate the x variable.

Step 2: Multiply the first equation by 3 (if you choose to eliminate y , then multiply the second equation by 2).

$$\begin{cases} 3x - 6y = 12 \\ 3x + y = 1 \end{cases}$$

Step 3: Now I can subtract the second equation and the x terms will cancel.

$$\begin{array}{rcl} 3x & - & 6y = 12 \\ -(3x & + & y = 1) \\ \hline 0 & - & 7y = 11 \end{array}$$

Step 3: Solve for y

$$y = -\frac{11}{7}$$

Step 4: Back-substitute into $x - 2y = 4$ to solve for x .

$$\begin{aligned} x - 2\left(-\frac{11}{7}\right) &= 4 \\ x + \frac{22}{7} &= 4 \\ x &= 4 - \frac{22}{7} = \frac{28}{7} - \frac{22}{7} = \frac{6}{7} \end{aligned}$$

So, the point $(\frac{6}{7}, -\frac{11}{7})$ is the solution to the system.

The next problem will showcase a different type of solution. And it has more variables. See what happens as you try the elimination method and why it might lead you to an interesting conclusion!

Problem 30. Solve the system using the elimination method. Notice that the second row is a multiple of the third row.

$$\begin{cases} x + y - z = -1 \\ 4x - 4y + 2z = 16 \\ 2x - 2y + 1z = 8 \end{cases}$$

Problem 31. A group of friends at the cafeteria bought 8 burritos, 6 cookies, and 6 fountain drinks for \$78.30. A second group of ordered 10 burritos, 6 cookies, and 8 fountain drinks for \$94.80. A third group ordered 3 burritos, 2 cookies, and 4 fountain drinks for \$32.85. Find a system of equations that allows you to solve for the price of each item, and then solve it.

2.3 The Matrix Method

Sometimes neither the substitution method nor the elimination method are very convenient. This is usually the case for a system with more than 3 equations or more than 3 variables. In this section, we learn a method of solving systems of equations that works well for pretty much any system. It is also easy to program into a computer and is therefore often used in the “real world.”

Definition 32. A *matrix* is a rectangular array of numbers, enclosed by brackets. An $n \times m$ matrix has n rows and m columns.

Example 33. A matrix with 2 rows and 3 columns (a 2×3 matrix): $\begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 5 \end{bmatrix}$.

Example 34. A matrix with 3 rows and 2 columns (a 3×2 matrix): $\begin{bmatrix} 1 & 0 \\ 3 & -1 \\ 5 & 2 \end{bmatrix}$.

We will use matrices to make it easier to solve a system of equations. First, we have to figure out how to transfer the information from the system into matrix form.

Problem 35. Can you think of a way to write the following system as a matrix?²

$$\begin{cases} x + 4y = 2 \\ 3x - y = 5 \end{cases}$$

The solution to this problem is a type of matrix called an **augmented** matrix. The matrix method of solving systems uses the same process as the elimination method, but now we don't have to keep writing all of the variables. There were three types of things we could do in the elimination method.

Problem 36. *In the elimination method, we usually focus on two equations at a time. It can be convenient to have these two equations isolated, or next to each other. In matrix form, we might also want to focus on certain rows or change the order of the rows. Switching the order of equations doesn't affect the answer in the elimination method. How would this work with a matrix representation of a system of equations? Show using an example.*

Problem 37. *In the elimination method, we could multiply an equation by a constant. This was helpful in allowing us to cancel variables. How would this work with a matrix representation of a system of equations? Show using an example.*

Problem 38. *In the elimination method, we could add or subtract equations (to cancel the variables). How would this work with a matrix representation of a system of equations? Show using an example.*

When using matrices, we are allowed to use these three types of operations, called row operations.

Definition 39. *A matrix is in **row echelon** form when the first nonzero entry in each row is a one and entries in the same column, below the one, are zeros. Any rows with all zeros should be at the bottom.*

Note: For an augmented matrix, we need only consider row echelon form for the left side of the augmented matrix!

Problem 40. *Are the following matrices in row echelon form (ref)? If not, then why?*

$$a) \left[\begin{array}{ccc|c} 1 & 3 & 2 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 7 \end{array} \right]$$

$$b) \left[\begin{array}{ccc|c} 1 & 3 & 2 & 5 \\ 0 & 1 & 4 & 6 \\ 2 & 0 & 1 & 7 \end{array} \right]$$

$$c) \left[\begin{array}{cc|c} 1 & 3 & 4 \\ 0 & 2 & 5 \\ 0 & 0 & 6 \end{array} \right]$$

$$d) \left[\begin{array}{cc|c} 1 & 3 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 6 \end{array} \right]$$

$$e) \left[\begin{array}{ccc|c} 1 & 0 & 5 & 4 \\ 0 & 1 & 2 & 4 \end{array} \right]$$

$$f) \left[\begin{array}{ccc|c} -1 & 0 & 5 & 4 \\ 0 & 1 & 2 & 4 \end{array} \right]$$

The advantage of row echelon form is how convenient it makes solving for the variables in a system of equations. You can see this in action through the next example.

Problem 41. Write the row echelon matrix as a system of equations. Then solve the system.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 4 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

At this point, let's be clear about what row operations are valid.

Matrix Method Row Operations:

Type 1: Interchange (switch) any two rows.

e.g. $R_1 \leftrightarrow R_2$.

Type 2: Replace a row by a nonzero multiple of that row.

e.g. $R_2 = 3r_2$.

Type 3: Replace a row by adding the constant multiple of another row. e.g. $R_2 = 3r_1 + r_2$.

Notation note: We will use capital R to denote the new row, which is found by performing an action, or row operation, to the lower case r old row. Also, the symbol $\leftrightarrow$ will be used when both rows are retained (hence the capital R) but the rows switch places.

Problem 42. Solve the system by using the matrix method to get row echelon form.

$$\begin{cases} 4x + 3y = 11 \\ x - 3y = -1 \end{cases}$$

- a) Write the system as an augmented matrix.
- b) Get a 1 in the upper left entry using a row operation.
- c) Get a 0 below that 1 using a row operation.
- d) Get a 1 to the right of the 0 using a row operation. You should now have a row echelon form matrix.
- e) Write the matrix as equations and solve the equations.

Definition 43. Solving a system of equations by using the matrix method to get row echelon form is called **Gaussian Elimination**.

Problem 44. Solve this system using the Elimination Method and Gaussian Elimination side by side. For each step of the elimination method, do the corresponding step of the matrix method next to it.

$$\begin{cases} -x + 2y = 3 \\ 4x + y = 1 \end{cases}$$

Recall that there are three possible types of solutions to a system of equations. It can have one solution (if the lines represented by the equations intersect), infinitely many solutions (lines are the same), or no solution (lines are parallel, but not the same).

Problem 45. Make up an example of a two-variable system with two parallel lines. Write the system as a matrix and get it into row echelon form. Based on the ref matrix, how can we tell in general if a system has no solutions, or in other words has parallel lines?

Problem 46. Follow the same steps as the previous problem, except start with a system that has coincident lines.

Problem 47. What is the solution to the following ref matrix?

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

2.4 Systems of m Linear Equations Containing n Variables

So far we have looked at systems with 2 or 3 variables and 2 or 3 equations, but there could be many more than that. For these more complicated systems, we might want to get the augmented matrix into an even simpler form so that we don't have to go through back-substitution. That form is called reduced row echelon form.

Definition 48. A matrix is in **reduced row echelon form (rref)** if

- The first nonzero entry in each row is a 1 and it has zeros above AND below it.
- The leading 1 in any row is to the right of the leading 1 in the row above it.
- Any rows that contain all zeros (on the left side of the vertical line) are at the bottom.

Problem 49. Are the following matrices in reduced row echelon form (rref)? If so, then what is the solution to the system of equations? If not, then why?

$$a) \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right]$$

$$b) \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$c) \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right]$$

$$d) \left[\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$e) \left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

$$f) \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \end{array} \right]$$

Notice how simple the left side of the matrices in parts *a* and *b* looks, with ones on the diagonal and zeros everywhere else. Matrices that look like those actually ends up being fairly important, as we'll see in later sections. They also has a specific name.

Definition 50. The **identity matrix**, I_n , is a square matrix (n rows and n columns) with ones on the diagonal and zeros otherwise.

$$\textbf{Example 51. } I_1 = [1] \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Problem 52. If you have an augmented matrix in rref and the left side looks like the identity matrix, how many solutions would the system have? What type of lines (parallel, intersecting, coincident) would be involved?

Definition 53. The process of solving a system of equations by getting a matrix into reduced row echelon form is called **Gauss-Jordan Elimination**.

Problem 54. Solve by getting the rref matrix (Gauss-Jordan elimination):

$$\begin{cases} x_1 + x_2 + x_3 = 4 \\ x_1 - x_2 - 9x_3 = 6 \end{cases}$$

Hint: your answer should have x_1 =an equation, x_2 =an equation, x_3 =any real number.

Problem 55. Solve by getting the rref matrix (Gauss-Jordan elimination):

$$\begin{cases} x - y + 2z = 2 \\ 2x - 3y + 2z = 1 \\ 3x - 5y + 2z = -3 \\ -4x + 12y + 8z = 10 \end{cases}$$

Problem 56. *You and your friends have a combined \$300 left on your meal plan and are all leaving campus tomorrow for summer break. As your money does not roll over to next semester, you want to spend all of it so that none of it goes to waste. You decide to buy bottled Starbucks coffee and Gatorade. You buy 11 oz. iced coffees for \$2, 9 oz. iced frappuccinos for \$3, and Gatorades for \$4. If you buy 600 oz. of coffee and 101 drinks, what did you buy? Solve using Gauss-Jordan elimination.*

Problem 57. *A toy company is analyzing sales reports for three products, xylophones (X), yo-yos (Y), and zebra figures (Z). One report shows that a combined total of 2000 of items X, Y, and Z were sold. Another report shows that the sum of the number of item Z sold and twice the number of item X sold equals 1000. Also, item X has 500 more items sold than item Y. Are these reports consistent?*

Chapter 3

Matrices

3.1 Matrix Algebra

Matrices don't always have to represent the coefficients in a system of equations. They are also useful to display a bunch of numbers, not necessarily coefficients. Sometimes we may need to add or multiply these matrices (depending on what the problem represents). Not every pair of matrices can be added or multiplied, though. It depends on the size of the matrix.

Definition 58. The *dimension* of a matrix is the number of rows, m , by the number of columns, n , and is denoted $m \times n$.

If $m = n$, then it is called a **square** matrix.

Problem 59. If you had to add the two matrices $\begin{bmatrix} 1 & 0 & 2 \\ 4 & -3 & 0 \end{bmatrix}$ and $\begin{bmatrix} 5 & -1 & 3 \\ 2 & 0 & -2 \end{bmatrix}$, what method would you use to do so? That is, how do you know which numbers to add together?

Problem 60. We cannot add any two given matrices. They have to fulfill a certain condition involving the dimension of each matrix. What do you think that condition is?

Problem 61. Find $\begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1 & 2 \\ -2 & 3 \end{bmatrix}$.

We can also multiply matrices by a scalar (a constant). See this in action in the next example, then apply it in the next problem.

Example 62. Multiply $\begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix}$ by 4.

Solution:

$$4 \begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix} = \begin{bmatrix} 4(1) & 4(-1) \\ 4(2) & 4(-2) \\ 4(3) & 4(-3) \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ 8 & -8 \\ 12 & -12 \end{bmatrix}$$

Problem 63. Given $A = \begin{bmatrix} 5 & -2 & 3 \\ 6 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 & 0 \\ 1 & -3 & 8 \end{bmatrix}$, find $2A - B$.

Problem 64. A nearby electronics store sells the following percentages of electronics on Black Friday:

Brand	Laptops	Cell Phones	TV	Tablet
Apple	15%	25%	5%	7%
Samsung	18%	22%	5%	3%

- Write a 2×4 matrix representing the counts of electronics
- On this given Black Friday, 500 units of electronics were sold. Determine the number of units sold of each type.

3.2 Matrix Multiplication

We'll start with just multiplying a 1-row matrix by a 1-column matrix.

Definition 65. If $R = \begin{bmatrix} r_1 & r_2 & \dots r_n \end{bmatrix}$ is a $1 \times n$ matrix and $C = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$ is an

$n \times 1$ matrix, then the product is

$$RC = \begin{bmatrix} r_1 & r_2 & \dots r_n \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = r_1c_1 + r_2c_2 + \dots r_nc_n$$

Example 66.

$$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = 0(3) + 1(4) + 2(5) = 0 + 4 + 10 = 14$$

What if I had two row matrices (i.e. $\begin{bmatrix} 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$)? I cannot multiply these two matrices. There is no definition for doing so.

Problem 67. Given the definition and example, what do you think a condition is for being able to multiply matrices?

Problem 68. Find RC if $R = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -3 \\ -1 \\ 3 \end{bmatrix}$.

Problem 69. Find RC if $R = [1]$ and $C = [2]$.

Definition 70. For $A = m \times r$ matrix and $B = r \times n$ matrix, the product AB is

$$AB = \begin{bmatrix} r_1 \text{ of } A \cdot c_1 \text{ of } B & r_1 \text{ of } A \cdot c_2 \text{ of } B & \cdots & r_1 \text{ of } A \cdot c_n \text{ of } B \\ r_2 \text{ of } A \cdot c_1 \text{ of } B & r_2 \text{ of } A \cdot c_2 \text{ of } B & \cdots & r_2 \text{ of } A \cdot c_n \text{ of } B \\ \vdots & & & \vdots \\ r_m \text{ of } A \cdot c_1 \text{ of } B & r_m \text{ of } A \cdot c_2 \text{ of } B & \cdots & r_m \text{ of } A \cdot c_n \text{ of } B \end{bmatrix}_{m \times n}$$

IMPORTANT NOTE: If A is 2×3 and B is 3×5 , then AB is 2×5 . The middle numbers must match to be able to multiply and the resulting product is the dimension of the outside numbers (2 and 5).

Problem 71. Let $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 1 \\ -4 & -3 & -2 \end{bmatrix}$. Find IA and

AI . I is the identity matrix. What do you notice about multiplying with I ?

Problem 72. Find AB if $A = \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 1 & -4 \end{bmatrix}$. Is this the same as BA ?

Problem 73. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, and $B = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$. First find AX . Then set AX equal to B . Solve for x and y .

Problem 74. Leonard went to a department store and purchased 6 pairs of pants, 8 shirts, and 2 jackets. Sheldon purchased 2 pairs of pants, 5 shirts, and 3 jackets. Set up a 2×3 matrix to represent this information, and label your rows and columns to identify what they represent. Assume that pants are \$25, shirts are \$18, and jackets are \$39. Represent that information as a matrix, and then use matrix multiplication to find the amounts spent by Leonard and Sheldon.

3.3 The Inverse of a Matrix

Before we get into the definition of a matrix inverse, it will help to discuss what the inverse of just a number is and why it is useful.

For example, what is the multiplicative inverse of the real number 2? The inverse is the number you multiply 2 by to get the identity, which is 1. One is called the identity because you can multiply any nonzero number by 1 and get the original number back. The multiplicative inverse of 2 is $\frac{1}{2}$ since $2 \cdot \frac{1}{2} = 1$ and $\frac{1}{2} \cdot 2 = 1$.

Why is this useful? You may not have realized it, but you often use multiplicative inverses when solving equations. If you were trying to solve $2x = 6$, what would you do? You would divide by 2, or multiply by $\frac{1}{2} = 2^{-1}$.

$$\begin{aligned} 2x &= 6 \\ \frac{1}{2}(2x) &= \frac{1}{2}(6) \\ x &= 3 \end{aligned}$$

Problem 75. What is the multiplicative identity for an $n \times n$ matrix?

Definition 76. Let A be an $n \times n$ matrix. The **inverse** of A , denoted A^{-1} , satisfies

$$AA^{-1} = A^{-1}A = I_n$$

NOTES:

- A and A^{-1} must be square matrices of the same dimension ($n \times n$). Otherwise, we wouldn't be able to find both AA^{-1} and $A^{-1}A$.
- The notation A^{-1} is used because of the connection to real numbers. Recall that the inverse of 2 is $\frac{1}{2}$ and fractions can be written using negative exponents, so $\frac{1}{2} = 2^{-1}$. This does NOT mean that we are dividing matrices as we do with real numbers. **You cannot divide by a matrix.**

Problem 77. Check that $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}$ is the inverse of $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$.

Finding the Inverse of an $n \times n$ Matrix, A :

- Step 1: Augment A with the appropriate identity matrix: $[A|I_n]$
(note: the right side is now a square matrix and not just a single column)
- Step 2: Use row operations to get the left side of the matrix in rref.
- Step 3: If the resulting augmented matrix looks like $[I|B]$,
then $B = A^{-1}$. Otherwise, A has no inverse.

Problem 78. Find the inverse of $A = \begin{bmatrix} 2 & 4 \\ 0 & -1 \end{bmatrix}$ and then check your answer.³

Problem 79. Find the inverse of $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$ and then check your answer.

Problem 80. Solve the system of equations $\begin{cases} 2x + 4y = 1 \\ -y = 2 \end{cases}$ using matrix inverses.

- Write the system as a matrix equation $AX = B$. The matrix A will have numbers, X will have variables, and B will be a column matrix with numbers.
- Find A^{-1} .
- Solve the matrix equation by multiplying both sides (on the left) by A^{-1} . Thus, you get

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B$$

$$X = A^{-1}B.$$

Chapter 4

Linear Programming: Graphical

Linear programming became a field of study after World War II. In that war, the U.S. Army needed to solve problems involving resource allocation. Resources, in that situation and in most other situations, are finite, leading to difficulties for the people in charge of distributing or using the resources.

4.1 Systems of Linear Inequalities

Definition 81. An *inequality* is an expression using $>$, $<$, $\geq$, or $\leq$. A *strict inequality* only uses either the symbol $<$ for “less than” or $>$ for “greater than.” Graphically, we represent a strict inequality with a dashed line. With graphs containing $\leq$ or $\geq$, a solid line is used.

Problem 82. Let x represent the amount of money required for the monthly utilities for an apartment and let y be the monthly rent. Represent, as an inequality, how much x and y can be if you allocate at most \$600 a month to these living expenses.

Problem 83. Graph $3x + 2y \geq 12$. Suppose x is the number of hours Alex spends writing a group paper and y is the number of hours Suzie spends on it. Come up with an example of what the 3, 2, and 12 might mean in this application example.

Problem 84. On separate graphs, draw

a) $y + 3 + x \leq -2$

b) $x > 2$

IMPORTANT NOTE: Remember that when you divide or multiply by a negative number, the inequality changes direction. For example, $-y \leq 3$ means $y \geq -3$ which should make sense because $-2 \leq 3$ and $2 \geq -3$.

Problem 85. Graph (on the same plane) the system of inequalities:

$$\begin{cases} -y \geq -x - 1 \\ y \geq -x + 1 \end{cases}$$

Problem 86. Give an example of a system of inequalities that has no solution. Your example can just be a graph.

Definition 87. A graph is called **unbounded** if the shaded areas go on forever.

Problem 88. You want to train for a triathlon and need to divide your time between swimming, biking, and running. You find a source online that tells you that newcomers to the sport should spend at least 80 minutes a week swimming, at least 150 minutes running, and at least 180 minutes biking. This source also provides two different daily training plans, which can be used individually or combined.

Plan A: 15 minutes swimming, 30 minutes running, 25 minutes biking

Plan B: 10 minutes swimming, 25 minutes running, 35 minutes biking

Write a system of inequalities that represent how many days of each plan should be done to reach the specified weekly minute goals. Graph your system to see possible training schedules.

4.2 A Geometric Approach for Problems with Two Variables

In the last section you learned about systems of inequalities. Now we are going to expand on that idea by looking at problems that combine inequalities with a special equation, called the objective function. We can solve these problems by using the graphing and shading methods learned previously.

Definition 89. A **linear programming problem** involves two things:

1. A linear objective function to be maximized or minimized.
2. A collection of linear inequalities that must be satisfied simultaneously.

If you were only given the constraints, or linear inequalities, then the problems would be the same as section 2.2. But now we also have a function that we are trying to maximize or minimize. This could be a profit function (maximize that!) or a cost function (minimize!), depending on the problem.

Problem 90. Farmer MacDonald raises goats and pigs. He wants no more than 16 animals, including no more than 10 goats. He spends \$25 to raise a goat and \$75 to raise a pig, and he has \$900 available. Find the maximum profit if each goat produces a profit of \$14 and each pig \$40.

- a) What could the variables x and y represent for this problem?
- b) What is the ultimate goal, or the main objective, for this problem? Write an equation to represent it.
- c) Write a system of inequalities to represent the total number of animals allowed and the total cost allowed. Graph this system.
- d) Your graph should have “corners”. What are these points? One of them will lead to the final answer. Thinking about your answer to part b, what should you do to see which corner point gives the answer? Also, what is the answer?

Problem 91. A school principal needs to replace some lockers in one of the hallways. CoolSkool lockers cost \$180 each, require 1.5 ft^2 of floor space, and hold 10 ft^3 of belongings. SpaceCase lockers cost \$120, require 2.5 ft^2 of floor space, and hold 14 ft^3 of belongings. The school has no more than \$2160 and 100 ft^2 of floor space. Maximize the amount of storage for belongings.

- a) What should the variables x and y represent for this problem?
- b) What is the ultimate goal, or the main objective, for this problem? Write an equation to represent it.
- c) Write a system of inequalities to represent the constraints that prevent the manager from getting whatever cabinets he wants. Then graph the system.
- d) Find and test the corner points and give the final answer.

Problem 92. Dante needs participants for a psychology study. He has a budget of \$500 for paying participants. People who participate in-person receive \$20 each but online participants only receive \$10. He wants to get at least 30 people to participate. How many in-person and online participants should he have to maximize participation?

- a) What should the variables x and y represent for this problem?
- b) What is the ultimate goal, or the main objective, for this problem? Write an equation to represent it.
- c) Write a system of inequalities to represent the constraints placed on Henry’s study. Graph the system.
- d) Find and test the corner points and give the final answer.

Chapter 5

Linear Programming: The Simplex Method

In the last chapter, you studied linear programming problems. You may have noticed that the examples in that chapter were relatively simple in that they only had 2 variables and the number of constraints was pretty small. What do we do if there are lots (thousands?) of variables or lots of constraints, as usually happens in the “real world”? Even though we would ideally want to use computers to solve such problems, understanding a more efficient strategy could reduce the computer’s running time from hours to minutes or seconds.

5.1 The Simplex Tableau: Pivoting

Problem 93. *If you had a linear programming problem with 4 variables and 8 constraints, how many corner points (intersections) could there be? How would you know which of these points are in the solution area (feasible region) and therefore should be checked with the objective function?*

What if we could find a way of only considering the feasible corner points? That would greatly reduce the number of points to test! In 1946, George Dantzig figured out such a strategy, called the simplex method.

Definition 94. *The **simplex method** is a strategy for solving linear programming problems by testing a solution at one corner point of the feasible set, then moving from there to another corner point at which the objective function has a better value, and so on until the best value is found.*

To start the simplex method, we need to know if a problem is set up a certain way.

Definition 95. *A maximum linear programming problem is in **Standard Form for a Maximum Problem** if:*

1. *All the variables are nonnegative.*

2. The other constraints can be written as a linear expression that is $\leq$ (positive constant).

Problem 96. Determine whether the following are in standard form. If not, why?

- a) Maximize $z = 2x_1 - x_2 + x_3$ subject to constraints:

$$\begin{aligned}x_1 + 5x_2 &\leq 6 \\x_2 - 3x_3 &\leq 20 \\x_1 + x_2 + x_3 &\leq 100 \\x_1 &\geq 0 \\x_2 &\geq 0 \\x_3 &\geq 0\end{aligned}$$

- b) Maximize $z = 5x_1 + 3x_3$ subject to constraints:

$$\begin{aligned}x_1 + x_2 - 5x_3 &\geq 5 \\x_1 - x_3 &\leq -7 \\x_1 + 2x_2 + 3x_3 &\leq 43 \\x_1 &\geq 0 \\x_3 &\geq 0\end{aligned}$$

Problem 97. Why do you think we need the variables to be nonnegative? Think about if this were a word problem.

Remember when we learned how to solve systems of equations that were too hard for the substitution or elimination methods? We used matrices!

Problem 98. How would you set up a matrix to represent a whole linear programming problem? You should think about what to do with the fact that these problems have inequalities instead of equations, and also have an objective function. Demonstrate your matrix using the standard form example above.⁴

Problem 99. Write the initial simplex tableau for the following linear programming problem:

$$\begin{aligned}\text{Maximize } z &= 3x_1 + 4x_2 \\ \text{subject to constraints:} \\ 2x_1 + 4x_2 &\leq 120 \\ 2x_1 + 2x_2 &\leq 80 \\ x_1 &\geq 0, x_2 \geq 0\end{aligned}$$

The next question is what to do with the simplex tableau once you find it. Remember that the simplex method goal is to find the best possible solution, or “corner point”, for the objective function. Also, remember how

we solved augmented matrices for systems of equations. We used row operations. Well, we want to use a specific set of those row operations for the simplex method. Keeping to these row ops will keep us in the feasible solution set of corner points. This process of particular row operations is called **pivoting**.

Definition 100. To *pivot* a matrix about a given element, called the **pivot element**, is to apply certain row operations so that the pivot element is replaced by a 1 and all other entries in the same column become 0s. The row containing the pivot element is called the **pivot row**.

Pivoting involves 3 steps:

1. Pick a pivot element. You will learn how to do this in a later section.
2. Make that pivot element a 1 using only the pivot row.
3. Make everything else in the pivot column a 0 (including the bottom row) by using the pivot row.

Problem 101. Given the initial simplex tableau

BV	P	x_1	x_2	s_1	s_2	s_3	RHS
s_1		0	2	7	1	0	50
s_2		0	4	1	0	1	10
s_3		0	2	2	0	0	20
P		1	-5	-8	0	0	0

The pivot element is 4 in row 2. What row operation will turn the 4 into a 1? What row operations can you use to get 0 everywhere else in the column? You do not have to actually do the row operations.

Problem 102. Given the linear programming problem to maximize $z = 4x_1 - x_2 + 3x_3$ with constraints

$$\begin{aligned}
 x_1 + x_2 + x_3 &\leq 100 \\
 2x_1 - 3x_2 + x_3 &\leq 50 \\
 -3x_1 + 2x_2 - x_3 &\leq 25 \\
 x_1 \geq 0, x_2 \geq 0, x_3 \geq 0
 \end{aligned}$$

- a) Find the initial simplex tableau. Show work for getting the constraints and objective equation in the correct equation format for the tableau (add slack variables, get objective =0).
- b) Pivot around the 2 in the second row, second column.

- c) What are the basic variables after you have pivoted? How do you know? Relabel all rows/columns.

Problem 103. Given the initial simplex tableau

BV	z	x_1	x_2	s_1	s_2	RHS
s_1	0	2	4	1	0	40
s_2	0	2	2	0	1	20
z	1	-3	-4	0	0	0

- a) Pivot using the number 2 in the 3rd column and the 2nd row as the pivot element.
- b) Write out the equation that each row represents. These are called the current equations for this problem. Set nonbasic variables equal to 0 to get the current solutions for the basic variables.

Definition 104. The **current value** of z (or any basic variable) is the value found by setting nonbasic variables equal to 0. For each equation that is represented by a row, this should result in a simplified equation where the BV labeled on the left is equal to the corresponding RHS value.

Note that this is not necessarily the final solution to z . It is only the current solution. Finding current solutions to a problem is called **analyzing** the tableau. It's kind of like you want to see what progress you've made, even though you aren't done with it.

5.2 Maximum Problems in Standard Form

Now you will learn how to pick the pivot element yourself.

It is not a random choice. The method for choosing the pivot element connects to the fact that we are dealing with maximizing an objective function.

Let me start with a given tableau.

BV	P	x_1	x_2	s_1	s_2	RHS
s_1	0	2	4	1	0	120
s_2	0	2	2	0	1	80
P	1	-3	-4	0	0	0

In this problem, we're trying to maximize $P = 3x_1 + 4x_2$. How?

If we can increase x_1 and x_2 as much as possible, then P will increase as much as possible.

Consider x_1 . How big can it be?

Problem 105.

- We know that $2x_1 + 4x_2 \leq 120$ by the first row of the tableau. If $x_2 = 0$, how big can x_1 be based on this inequality?*
- We know that $2x_1 + 2x_2 \leq 80$ by the second row of the tableau. If $x_2 = 0$, how big can x_1 be based on this inequality?*
- Now consider both inequalities. What is the largest that x_1 can be? Whichever row gave us this number is the one we want to pivot, with the pivot column being the x_1 column. If we use that row, we will stay within the feasible solution for x_1 .*
- What happens if you choose the wrong row to pivot? Try it and see.*

The following summarizes what we just explored, and also tells you how to choose which column to focus on.

METHOD FOR CHOOSING THE PIVOT ELEMENT	
Step 1:	Find the column that contains the first negative number in the objective row (last row).
Step 2:	For every positive number in that column, find the ratio of the RHS to that number.
Step 3:	Whichever ratio is smallest, use that row as the pivot row and that entry as the pivot element.

Problem 106. Find the pivot element of each tableau:

a)

BV	P	x_1	x_2	s_1	s_2	RHS
s_1	0	2	3	1	0	50
s_2	0	5	8	0	1	10
P	1	-6	-2	0	0	0

b)

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
s_1	0	2	3	1	0	50
s_2	0	5	8	0	1	10
<i>P</i>	1	6	-2	0	0	0

c)

<i>BV</i>	<i>P</i>	x_1	x_2	x_3	s_1	s_2	<i>RHS</i>
s_1	0	1	2	3	1	0	25
s_2	0	9	0	4	0	1	14
<i>P</i>	1	5	-1	-2	0	0	0

d)

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
s_1	0	3	-2	1	0	0
s_2	0	2	0	0	1	50
<i>P</i>	1	1	-3	0	0	0

Problem 107. Find the pivot element of the following tableau and then pivot. What is the current solution?

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
s_1	0	0	2	1	-1	40
x_1	0	1	1	0	$\frac{1}{2}$	40
<i>P</i>	1	0	-1	0	$\frac{3}{2}$	120

Problem 108. How do you know when to stop pivoting? In other words, how do you know when the current solution is the final solution? Think about what it will mean for the objective row.

Problem 109. Decide if the tableau is final. If it is, give the solution. If not, pick the pivot element. You do not have to actually pivot.

a)

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
x_1	0	1	0	-1	0	21
x_2	0	0	1	-1	1	34
<i>P</i>	1	0	0	7	4	90

b)

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
x_1	0	1	4	0	0	21
s_1	0	0	-4	1	1	34
<i>P</i>	1	0	3	0	4	90

c)

<i>BV</i>	<i>P</i>	x_1	x_2	s_1	s_2	<i>RHS</i>
x_1	0	1	4	0	0	20
s_1	0	0	4	1	1	20
<i>P</i>	1	0	-3	0	4	10

Problem 110. Create a chart describing the simplex method. Label each step and write a brief description of what to do in that step. This chart is based on being given the original linear programming problem (not as a matrix) and going through each step until you have a final solution.

Problem 111. Find the solution of the linear programming problem.

$$\text{Maximize } P = 5x_1 + 4x_2$$

subject to constraints:

$$x_1 + x_2 \leq 2$$

$$2x_1 + 3x_2 \leq 6$$

$$x_1 \geq 0, x_2 \geq 0$$

Problem 112. Pam's tree service specializes in trimming trees. Pam charges different amounts for trimming trees based upon the height of each tree. She charges \$300 for trees less than 30 feet tall, \$525 for trees between 30 and 60 feet tall, and \$800 for any tree over 60 feet tall. Pam has also found that on average it takes her crew two hours to complete a job on the shorter trees, four hours for medium height trees, and eight hours for tall trees. Pam has four crews that can each work 40 hours/week. Additionally, Pam's tree service can work on at most 30 jobs per week. Pam wants to know how many of each size tree she should attempt to schedule each week in order to maximize revenue.

a) Formulate a linear programming problem that models the problem given above. Be sure to identify all variables.

b) Solve the linear programming problem.

5.3 The Duality Principle for Minimum Problems

A minimum problem is when you are trying to minimize an objective function instead of maximizing it. Often the objective function represents cost. We like to minimize how much money is spent, for example.

Standard form for a minimum problem is NOT the same as it was for a maximum problem. You need to know the difference between them.

Definition 113. *The standard form of a minimum problem must satisfy:*

1. All variables are nonnegative.
2. All other constraints are $\geq$ a constant.
3. The objective function has nonnegative coefficients.

Problem 114. *Describe all the differences between the standard form definitions for max and min problems.*

Problem 115. *Is the problem in standard form? If it is not, explain why. Can we fix it so that it is standard form?*

a) Minimize $C = x + 5y$ subject to constraints:

$$\begin{aligned} 2x - y &\geq 4 \\ 3x + y &\geq 1 \\ x \geq 0, y &\geq 0 \end{aligned}$$

b) Minimize $C = -x + 5y$ subject to constraints:

$$\begin{aligned} 2x - y &\leq 4 \\ 3x + y &\geq 1 \\ x &\geq 0 \end{aligned}$$

What we want to do first is transform the minimum problem into a maximum problem. We know how to solve a SF max problem now: Set up tableau and pivot until no negatives in last row. How can we turn a min problem into a max problem?

Answer: Find the dual problem.

THE DUALITY PRINCIPLE: If the minimum problem has a solution, the solution equals the maximum value of the dual problem.

You have not yet seen a dual problem, but this principle lets you know that if you can find the dual problem, then go ahead and solve it as a maximum problem and you actually end up with the solution to the original min problem.

Before a demonstration of how to find the dual problem, you need to know what the transpose of a matrix is.

How to Take the Transpose of a Matrix:

- If you have a $m \times n$ dimensional matrix (m rows and n columns) then the transpose will be $n \times m$.
- The first row in the original matrix becomes the first column in the transpose.
- The second row in the original matrix becomes the second column in the transpose.
- Eventually you switch all of your rows into columns and you have the transpose!

Problem 116. *Take the transpose of the matrix.*

$$a) \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$b) \begin{bmatrix} 0 & 3 & -4 \\ 1 & -2 & 3 \\ -1 & 5 & 10 \\ 9 & 7 & 8 \end{bmatrix}$$

Method	to Find the Dual Problem of a Minimum Problem:
Step 1:	Make sure it is in standard form. If not, can you multiply a constraint by a negative to get it in SF?
Step 2:	Write the minimum problem as an augmented matrix.
Step 3:	Take the transpose.
Step 4:	Write the dual maximum problem (include the word maximum, use new variables, and don't forget the nonnegative constraints)

Example 117. *Find the dual problem and first pivot element of the linear programming problem.*

$$\begin{aligned} \text{Minimize } C &= 10x_1 + 11x_2 \\ \text{subject to constraints:} \\ -x_1 - 2x_2 &\leq -3 \end{aligned}$$

$$\begin{aligned}4x_1 + 5x_2 &\geq 6 \\7x_1 + 8x_2 &\geq 9 \\x_1 \geq 0, x_2 &\geq 0\end{aligned}$$

Solution:

This problem is not in standard form since the first constraint has $\leq$ instead of $\geq$.

Multiply the constraint by a negative 1 to fix this.

$$-(-x_1 - 2x_2) \geq -(-3) \Rightarrow x_1 + 2x_2 \geq 3$$

Set up an augmented matrix

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 0 \end{array} \right]$$

Take the transpose.

$$\left[\begin{array}{ccc|c} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 0 \end{array} \right]$$

The dual problem is

$$\text{Maximize } P = 3y_1 + 6y_2 + 9y_3$$

subject to constraints:

$$y_1 + 4y_2 + 7y_3 \leq 10$$

$$2y_1 + 5y_2 + 8y_3 \leq 11$$

$$y_1 \geq 0, y_2 \geq 0, y_3 \geq 0$$

Now solve this problem using the simplex method.

Initial tableau:

<i>BV</i>	<i>P</i>	<i>y₁</i>	<i>y₂</i>	<i>y₃</i>	<i>s₁</i>	<i>s₂</i>	<i>RHS</i>
<i>s₁</i>	0	1	4	7	1	0	10
<i>s₂</i>	0	2	5	8	0	1	11
<i>P</i>	1	-3	-6	-9	0	0	0

Pick the pivot element.

y₁ column since it has the first negative in the P row.

$$\frac{10}{1} = 10 \text{ and } \frac{11}{2} = 5.5$$

5.5 < 10 so pick pivot element 2 in y₁ column and s₂ row.

Then pivot...(I'm not going to write out the rest of the steps)

Important Notes:

- The duality principle tells us that the solution to P in the dual problem is also the solution to the original (primal) minimum problem. The solutions for $x_1, x_2, \dots$ are slightly different (but easy!).
- You CANNOT use x as your variable in the dual problem, since it won't represent the same number as the primal problem x . It needs to be different. I will use y .
- Don't forget to switch the constraint inequalities to $\leq$ for the dual problem.
- The last row of the augmented matrix should be the objective function, both before and after transposing.

Problem 118. When we solve a dual problem, we have y_i variables instead of the original x_i 's. We want to solve for C and x_i , though. If a primal problem has x_1, x_2 , and x_3 , then how many constraints (ignore the ≥ 0 constraints) will the dual problem have? How many slack variables will the dual problem have? Based on your answers, guess how we solve for x_i .

Problem 119. Solve the minimum problem.

$$\begin{aligned} \text{Minimize } C &= 3x_1 + 2x_2 \\ \text{subject to constraints:} \\ x_1 + 3x_2 &\geq 6 \\ 2x_1 + x_2 &\geq 3 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

Problem 120. Create a chart describing the simplex method for standard form minimum problems. Label each step and write a brief description of what to do in that step. This chart is based on being given the original linear programming problem (not as a matrix) and going through each step until you have a final solution.

Problem 121. Find the dual problem (do not solve it) to the minimization problem

$$\begin{aligned} \text{Minimize } C &= 5x_1 + 3x_2 + 2x_3 \\ \text{subject to constraints} \\ 2x_1 - 3x_2 + x_3 &\geq 4 \\ -x_1 + 3x_2 + 2x_3 &\geq 1 \\ x_1 \geq 0, x_2 \geq 0, x_3 &\geq 0 \end{aligned}$$

Problem 122. What is the solution to the primal problem if the final tableau

of the dual problem is

<i>BV</i>	<i>P</i>	<i>y</i> ₁	<i>y</i> ₂	<i>y</i> ₃	<i>s</i> ₁	<i>s</i> ₂	<i>RHS</i>
<i>s</i> ₁	0	0	2	5	1	8	11
<i>y</i> ₁	0	1	3	6	0	9	12
<i>P</i>	1	0	4	7	0	10	13

5.4 Non-Standard Form Problems

First we'll look at non-SF maximum problems.

What if the only reason that a maximum problem is not in SF is because a constraint has the wrong inequality? For example, one constraint could be $x_1 + x_2 \geq 7$. When this occurred for minimum problems, we tried multiplying the whole thing by -1. Let's do that here:

$$\begin{aligned} -(x_1 + x_2) &\leq -7 \\ -x_1 - x_2 &\leq -7 \end{aligned}$$

Now the inequality is the right one, but it is not $\leq$ a **positive** constant, so this is still not a SF maximum problem. We need a different method for solving this.

Use an alternate pivoting strategy if the right hand side (RHS) of the tableau has a negative entry. To select the pivot row look for the most negative entry in the RHS column. By most negative I mean if you draw a number line, it would be the number farthest to the left. Or, in other words, the smallest number. -10 is "more" negative than -5.

Alternate Pivoting Strategy (APS)

- | | |
|----------|---|
| Step 1: | Look at the RHS column and consider the row with the biggest negative entry in that column
(e.g. -12 is 'bigger' than -11 in the example below). |
| Step 2a: | If there is a negative on the left side in the row chosen in Step 1, pick the left-most negative number in the row as the pivot element. |
| Step 2b: | If there are no other negatives in the row, the problem has no solution. |

Problem 123. Pick the first pivot element using the alternate pivoting strategy.

a)

<i>BV</i>	<i>P</i>	x_1	x_2	x_3	s_1	s_2	<i>RHS</i>
s_1	0	1	2	5	1	0	-11
s_2	0	0	-3	-6	0	1	-12
P	1	-5	-4	-7	0	0	0

b)

<i>BV</i>	<i>P</i>	x_1	x_2	x_3	s_1	s_2	<i>RHS</i>
s_1	0	1	2	5	1	0	-11
s_2	0	0	-3	-6	0	1	12
P	1	-5	-4	-7	0	0	0

Problem 124. Solve the linear programming problem.

$$\begin{aligned}
 &\text{Maximize } P = 4x_1 + x_2 \\
 &\text{subject to constraints} \\
 &\quad x_1 + x_2 \geq 2 \\
 &\quad 2x_1 + x_2 \leq 8 \\
 &\quad x_1 \geq 0, x_2 \geq 0
 \end{aligned}$$

If a minimum problem is not in SF, I want you to use a different method than finding the dual as we did before. Now that we know the alternate pivoting strategy for non-SF max problems, we can also use that to help with non-SF min problems.

Problem 125. Motivating our new min-to-max method: Suppose that $40 \leq C \leq 100$. What is the minimum of C ? If we want to consider $P = -C$, what is the maximum of P ?

We can turn a min problem into a max problem just by taking the negative of it.

Problem 126.

$$\begin{aligned}
 &\text{Minimize } C = 5x_1 + 6x_2 \\
 &\text{subject to constraints} \\
 &\quad x_1 + x_2 \leq 10 \\
 &\quad x_1 + 2x_2 \geq 12 \\
 &\quad 2x_1 + x_2 \geq 12 \\
 &\quad x_1 \geq 3, x_2 \geq 0
 \end{aligned}$$

a) Why is this min problem not in standard form?

- b) Write the min problem as a max problem where $P = -C$. You will want to change the inequalities to be $\leq$, as we like for a max problem.
- c) Write the initial simplex tableau for the max problem $P = -C$.
- d) Pick the first pivot element.
- e) The final tableau for this problem is

BV	P	x_1	x_2	s_1	s_2	s_3	RHS
x_2	0	0	1	0	$-\frac{2}{3}$	0	4
x_1	0	1	0	0	$\frac{1}{3}$	0	4
s_3	0	0	0	1	$\frac{1}{3}$	1	2
P	1	0	0	6	$\frac{7}{3}$	0	-44

What do you think the solution to the original problem is?

Important Notes:

- Notice in the last example that the objective row in the initial tableau had no negative numbers but it was still not a final tableau since the alternate pivoting strategy was needed!
- When solving a $P = -C$ problem, your final tableau will have a negative number in the bottom right spot (or the entry that is the solution to P). Since you take the negative of this, you end up with a positive value of C .

The last type of non-SF problem we'll look at is a linear programming problem, max or min, that has a constraint with an equal sign ($=$) instead of an inequality.

Problem 127. Suppose you are considering the problem:

$$\begin{aligned}
 &\text{Minimize } C = 7x_1 + 5x_2 + 6x_3 \\
 &\text{subject to constraints} \\
 &\quad x_1 + x_2 + x_3 = 10 \\
 &\quad x_1 + 2x_2 + 3x_3 \leq 19 \\
 &\quad 2x_1 + 3x_2 \geq 21 \\
 &\quad x_1 \geq 0, x_2 \geq 0, x_3 \geq 0
 \end{aligned}$$

- a) Why is this not standard form?
- b) Start by creating the $P = -C$ max problem. Leave the $=$ as is.

- c) Set up the initial simplex tableau. What do you think happens to the equality constraint? Consider what the purpose of slack variables is.*
- d) How can we get a replacement basic variable?*
- e) Pick a pivot element and then pivot.*

Problem 128. *Create a chart describing the method of solving linear programming problems with equality constraints.*

Chapter 6

Sets and Counting

6.1 Sets

You have probably heard the term “set” before. Maybe in reference to a set of numbers or a set of objects. Here’s the formal definition.

Definition 129. A *set* is a collection of distinct objects or elements.

The word distinct in that definition means different. Each object is different, or at least you don’t list repetitions when describing the set.

Example 130.

- The planets are a set: $\{\text{Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune}\}$.
- The integers are a set of numbers: $\{0, \pm 1, \pm 2, \dots\}$.
- The even integers are also a set of numbers: $\{0, \pm 2, \pm 4, \dots\}$.
- The collection of numbers $\{1, 2, 2\}$ is NOT a set since 2 is repeated. The elements are not distinct.

Problem 131. Give an example of...

- a) A set of letters.
- b) A set of objects in the room you are in.
- c) A set of something not in the room.

Definition 132. The *empty set* is the set of nothing, or the set that is empty (denoted $\emptyset = \{ \}$).

Sometimes we want to use just part of a set. There is a definition for this.

Definition 133. Let A and B be sets. A is a *subset* of B if every element in A is also in B .

- A being a subset of B is denoted $A \subseteq B$.
- This notation can also be read as “ A is contained in B ”.

Problem 134. Let S be any set other than the empty set. We know that S has at least two subsets. What are they?

Problem 135. Find all subsets of $A = \{1, 2, 3\}$.

If we know that A cannot be equal to B , then we can be more specific.

Definition 136. A is a **proper subset** of B , denoted $A \subset B$, if A is a subset of B and A is not equal to B .

Hopefully, that notation makes sense. It is similar to an inequality $\leq$ versus $<$. The former means it can be equal to and the latter it cannot be equal. We leave off the partial equal sign for proper subsets, also.

Problem 137. Give an example of...

- A subset of the even integers.
- A proper subset of the set $A = \{1, 2\}$.
- A proper subset of the set $A = \{a\}$

Problem 138. Decide whether the following are true or false. Justify your answer.

- $\{1, 2\} \subseteq \{1, 3, 4\}$
- $\{x, y, z\} \subseteq \{x, y, z\}$

Definition 139. The **universal set**, U , is the set consisting of all elements under consideration (i.e. the set of everything).

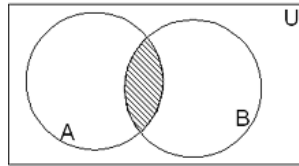
Example 140. Suppose you were considering a set $S = \{\text{Finite Math, Calculus}\}$. Then the universal set might be all math classes offered at our university.

Sometimes we want to know what two sets have in common. For example, if one set was items on sale at a store and another set was items you have a coupon for, you might want to know what items are on sale *and* you have a coupon for.

Definition 141. The **intersection** of sets A and B , written $A \cap B$, is the set $A \cap B = \{x \mid x \text{ is in } A \text{ and } B\}$ (read as “ x such that x is in A and B ”).

The intersection of A and B is represented by the shaded area in the following Venn diagram:

We also need a name for the elements that are in *either* A or B . By this I mean any element that is in A , in B , or in both.



Definition 142. The **union** of A and B , denoted $A \cup B$, is the set $A \cup B = \{x \mid x \text{ is in } A \text{ or in } B \text{ (or both)}\}$.

Problem 143. Draw a Venn diagram that describes the union of:

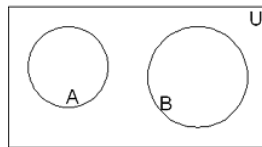
- a) two sets, A and B
- b) three sets, A , B , and C

Problem 144. Given $A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$, and $C = \{3, 4, 5\}$, what is

- a) $A \cup B$?
- b) $A \cap C$?
- c) $A \cup (B \cap C)$?

What if A and B don't have any elements in common? In that case, the intersection would be the empty set, or $A \cap B = \emptyset$. We have a name for this, too.

Definition 145. If A and B share no elements, then $A \cap B = \emptyset$ (the empty set) and A and B are **disjoint**.



Definition 146. Let A be any set. The **complement** of A , denoted $\bar{A}$, is the set $\bar{A} = \{x \mid x \text{ is not in } A\}$.

Problem 147. Draw a Venn diagram of the complement of a set A .

Problem 148. Describe:

- a) $A \cap \bar{A}$
- b) $A \cup \bar{A}$
- c) $\overline{\bar{A}}$

Problem 149. Suppose a 6-sided die is tossed. Consider the possible outcomes.

- What is the universal set of outcomes?
- Find $A = \{?\}$, the set of outcomes in which an even number appears.
- Find $B = \{?\}$, the set of odd outcomes.
- Are A and B disjoint?
- Find the complement of A .

6.2 The Number of Elements in a Set

For any set A , let $n(A)$ be the number of elements in A .

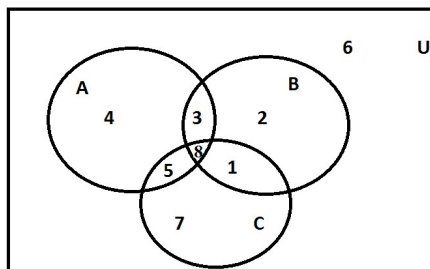
Problem 150. Find $n(A)$ for the following:

- $A = \text{the letters in the English alphabet (a through z)}$.
- $A = \emptyset$.
- $A = \{\text{Dr. Capaldi}\}$

We have a formula for counting sets that is very useful.

Theorem 151. Counting Formula: Given sets A and B , $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Problem 152. The numbers in this Venn diagram represent how many elements are in that part of the circle or space of the diagram.



- Find $n(A)$.
- Find $n(B)$.
- Find $n(A \cap B)$.

- d) Find $n(A \cap B \cap C)$.
- e) Find $n(A \cup B)$ using the counting formula above. Then verify by checking the Venn diagram.

Problem 153. Suppose you have two sets, E and F . Give formulas for the following based on the Counting Formula:

- a) $n(E \cap F) = \dots$ (formula in terms of $n(E)$, $n(F)$, and/or $n(E \cup F)$)
- b) $n(F) = \dots$ (formula in terms of $n(E)$, $n(F)$, and/or $n(E \cup F)$)
- c) $n(\bar{E}) = \dots$ (formula in terms of $n(E)$, $n(F)$, $n(U)$, and/or $n(E \cup F)$)

Problem 154. According to the American Psychological Association, based on 13,984 worldwide college students who were surveyed between 2014 and 2017, 35% of respondents reported experiencing a mental disorder sometime in their life. Thirty-one percent reported experiencing a mental disorder in the last 12 months. Draw a Venn diagram representing the numbers of students for the lifetime and 12 month sets.

Source: <https://www.apa.org/pubs/journals/releases/abn-abn0000362.pdf>

Problem 155. Suppose that in a survey of 500 CEOs, 310 had degrees in business. Of these, 238 had an undergraduate business degree and 184 had graduate business degrees. Use the counting formula to find how many had both.

6.3 The Multiplication Principle

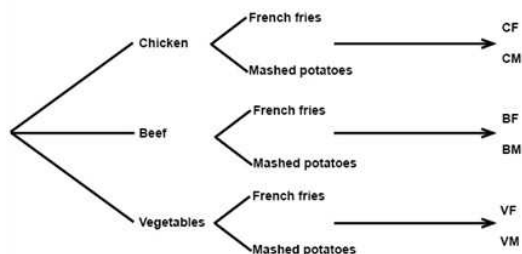
Example: A wedding reception dinner gives guests two options: the main entree and the side dish. There are three entree options: chicken, beef, or vegetarian. The two side options are french fries or mashed potatoes. How many dinner options does a guest have?

Often, problems like this are useful to demonstrate with tree diagrams. You can see from this diagram that there are 6 options.

Problem 156. Draw a new tree diagram for the last example, but with the side options listed first and the entree options second.

- a) Does reordering the tree change the answer?
- b) How do we solve this without drawing a diagram?

Problem 157. A health insurance provider offers 10 medical, 5 dental, and 4 vision plans. How many different insurance combinations are available if someone wishes to purchase all three types of coverage?



Problem 158. Finish the following statement.

Multiplication Principle of Counting: If a task consists of a sequence of choices in which there are p_1 number of selections for the first choice, p_2 selections for the second choice, p_3 selections for the third choice, and so on, then the total number of ways these selections can be made is...

Problem 159. A Social Security number has 9 digits. How many Social Security numbers are possible if the only restriction is that the number 000-00-0000 cannot be assigned? The U.S. population in 2012 was approximately 313 million. Do you think the U.S. is in danger of running out of Social Security numbers?

Problem 160. Until recently, all area codes for phone numbers had a 0 or a 1 as the middle digit and the first digit could not be 0 or 1. How many area codes were possible with this arrangement?

Problem 161. How many 4-digit numbers can be formed using the digits $\{1, 2, 3, 4, 5, 6, 7\}$ if adjacent digits must be different?

6.4 Permutations

In the last two sections of this chapter we will learn about two counting techniques that can be really helpful when solving probabilities, which come up in the next chapter. These topics should remind you of the multiplication

principle.

The first type of counting problem has to do with permutations, which involve rearranging elements of a set into some order (permuting). There are two types of permutations.

Problem 162. *The IT department is making you change your password. You need to come up with a new one that has 6 letters. This IT department is relaxed and does not require a ‘strong’ password, so there are no requirements other than it including only letters and being length 6.*

- a) Are the letters you can choose from distinct (i.e. different or unique)?*
- b) Is repetition of the letters allowed?*
- c) Is the order of the letters important?*
- d) How many possible passwords are there?*

Definition 163. *Consider r objects selected from n distinct objects where*

- $r \leq n$
- *the n objects are distinct*
- *objects can be repeated*
- *order is important*

The formula for counting (repetition allowed) permutations is _____.

We saw some multiplication principle problems, where repetition was not allowed, the number of options for each selection decreased after the previous selection was made. To have a nice formula for these types of permutations, I need to introduce some new notation.

Definition 164. Factorial

*The symbol $n!$, read as “**n factorial**”, is defined as:*

$$\begin{aligned}
 0! &= 1 \\
 1! &= 1 \\
 2! &= 2 \cdot 1 \\
 3! &= 3 \cdot 2 \cdot 1 \\
 4! &= 4 \cdot 3 \cdot 2 \cdot 1 \\
 &\vdots \\
 n! &= n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 3 \cdot 2 \cdot 1
 \end{aligned}$$

Problem 165. Calculate the following factorials.

- a) $6!$
- b) $\frac{5!}{4!}$
- c) $\frac{7!}{(7-5)!5!}$
- d) $\frac{75!}{70!}$
- e) $\frac{12!}{6!6!}$

So, when do we need to use factorials to help count the number of permutations for a problem? Consider the following example.

Problem 166. Given the set $\{a, b, c, d, e\}$:

- a) How many arrangements of 3 letters can be formed from this set if repetition is allowed?
- b) What if the letters $\{a, b, c, d, e\}$ represent people? How many lines of 3 people can be formed?
- c) Your answer to part (b) should resemble a factorial except the last few numbers (say, r numbers) are left off. Try to write your answer as a factorial divided by another factorial (to cancel the r numbers).

The following definition generalizes your answer to the previous problem.

Definition 167. Consider r objects selected from n distinct objects where

- $r \leq n$
- the n objects are distinct

- *an object once used cannot be repeated*
- *order is important*

The formula for counting (repetition not allowed) permutations is $P(n, r) = \underline{\hspace{2cm}}$.

Problem 168. *Four candidates will be elected as president, vice president, treasurer, and secretary of a club with 20 people in it.*

- Are the people you can choose from distinct?*
- Is repetition of the people allowed?*
- Is the order that the people are chosen important?*
- How many possible outcomes are there?*

Problem 169. *Eight horses are entered in a race which has 1st, 2nd, and 3rd place prizes. Assuming there are no ties, how many outcomes for the prizes are possible?*

Problem 170. *To play the Lucky 8 lottery, suppose you have to choose three numbers between 1 and 8, repetition allowed. How many possible lottery tickets are there?*

6.5 Combinations

Definition 171. *A **combination** is an arrangement of r objects selected from n distinct objects where*

- $r \leq n$
- *The n objects are distinct.*
- *Repetition is not allowed.*
- *Order does not matter.*

The formula for counting combinations is $C(n, r) = \frac{n!}{r!(n-r)!}$.

Important Note: The formula $C(n, r)$ has another notation that is commonly used, $\binom{n}{r}$, which is read as “ n choose r ”. Hopefully, that makes sense be-

cause we are choosing r number of objects from n number of objects.

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Problem 172. From the four numbers $\{1, 2, 3, 4\}$, how many ways are there to choose three without repeating any number...

- a) if order is important.
- b) if order is not important.
- c) Based on your answers to parts (a) and (b), why do you think we need the extra $r!$ in the denominator for the combinations formula?

Problem 173. For each problem, decide whether order is important, whether there can be repetition, and count the number of possible arrangements.

- a) A board of directors with 10 members needs to choose 3 for a committee. How many possible committees can there be?
- b) 10 books are being proposed for a book club. How many ways can the top three book choices (1st, 2nd, 3rd) be filled?
- c) You have 9 sweaters. Your suitcase can only fit 4. How many ways can you select 4 of them to take on a trip?

Problem 174. You buy 12 apples from County Line Orchard. Once you finished eating the apples, it ended up that 2 were bad and the other 10 good.

- a) If you had randomly selected 5 of those apples to eat, how many samples (groups of 5) would have all good apples?
- b) If you had randomly selected 5 of those apples to eat, how many samples (groups of 5) would have 1 bad apple and 4 good?

Problem 175. A new university gaming club has been formed. From a group of 4 women and 6 men, an e-board of 3 members must include at least one woman. How many ways can this happen?

- a) First decide on order and repetition.
- b) Now split the problem up into three cases/events and find how many ways the e-board can be chosen for each case.
 - Case 1: 1 woman and 2 men
 - Case 2: 2 women and 1 man
 - Case 3: 3 women and 0 men

c) How do you combine the cases? What is the final answer once the cases are combined?

Problem 176. *In general, when would you use addition in a counting problem?*

Chapter 7

Probability

7.1 Sample Spaces and Probabilities

There are certain events in the real world that will not always produce the same outcome. A few examples:

- Matches between sports teams.
- Playing cards.
- The weather.

What we want to do in these situations/experiments is figure out the **probability** that an outcome will occur in the long run.

In order to find a probability, we need to know how many possible outcomes there are. The name for the set of all possible outcomes is the sample space. It is a similar idea to the universal set.

Definition 177. A *sample space*, $S = \{e_1, e_2, \dots, e_n\}$, of an experiment is the set of all possibilities that can occur as a result of the experiment. Each element, e_i , of a sample space is called an **outcome**.

Problem 178. Suppose Jimmy flips a coin three times. Assume that heads-heads-tails counts as a different outcome than tails-heads-heads.

- a) What is the sample space for this experiment? Actually write out each outcome in set notation.
- b) What is the probability of each outcome? This is called the **probability model** for the experiment.
- c) What is the probability of the sample space?

Problem 179. Get a 6-sided die. ⁵

- a) What is the probability model for rolling the die once?

- b) Roll the die until you get a 3 and mark down how many rolls it took. Based on your answer to the first question, does the probability of getting a 3 make sense given how many people in the room got a 3 on the first roll?
- c) How many outcomes are in the sample space for rolling a die twice (can you make a tree diagram)? What is the probability of getting two 3's in a row? Roll the die twice and we will see how many people get two 3's.

Definition 180. An *event*, E , is any subset of the sample space.

Example 181. In the last problem, we used the event $E = \{3\}$ from the sample space where the die was rolled once. For rolling twice, the event was $E = \{(3, 3)\}$. You could also consider an event of the set of even numbers on the die. Then the probability of $E = \{2, 4, 6\}$, or $P(E)$, would be the probability of you rolling an even number.

Problem 182. What is the probability of rolling an even number if you roll the die once? Try rolling it 6 times. How many of those were even? Is the probability of getting an even number different or the same on the first roll versus the 6th roll?

Problem 183. Suppose the number of outcomes in a sample space is $n(S)$ and the number of outcomes in an event is $n(E)$. What is a formula for the probability of E (notation $P(E)$)?

Problem 184. You would never do this, but your friend wants to cheat by using a “loaded” die. It will land on the number 2 ten times as often as the other numbers. If the die is rolled once, what is the probability of each outcome in the sample space?

Problem 185. Look around the room and count up how many people are wearing glasses. If a person were randomly chosen from the room, what is the probability that they are wearing glasses (i.e. if $E = \{\text{wearing glasses}\}$, find $P(E)$)? What is the probability of the complement, $P(\bar{E})$, that they are not wearing glasses?

Problem 186. Is it possible to have a negative probability? Why or why not?

Problem 187. What do you think is the largest possible probability for any event? Why?

Problem 188. American Public Media (apmreports.org) investigated the racial composition of juries for trials of Curtis Flowers, a black man who has been tried six times for the same crime. In his latest trial, there were 56 potential jurors. Fifteen of those people were black and 41 were white. The final jury had 1 black member and 11 white members. What is the probability, if jurors were selected randomly, of ending up with this jury?

7.2 Properties of the Probability of an Event

In this section we will compile a list of useful properties of probabilities. The first one was actually touched on in the last section.

Theorem 189. *If S is a sample space such that $S = \{e_1, e_2, e_3, \dots, e_m\}$, then*

$$P(S) = P(e_1) + P(e_2) + P(e_3) + \cdots + P(e_m) = 1$$

This property basically says that there is 100% chance that something will happen. Out of all possible outcomes, one of them must occur.

These individual outcomes, e_i , are called simple events. They are simple because each of them is only one outcome. We will assume that all simple events are equally likely.

A regular event is some combination of one or more simple events, like $E = \{e_1, e_3\}$.

Theorem 190. *If E is an event such that $E = \{e_1, e_2, e_3, \dots, e_n\}$, then*

$$P(E) = P(e_1) + P(e_2) + P(e_3) + \cdots + P(e_n)$$

Problem 191. *The number of cars going through the local roundabout is recorded from 5 to 6 PM for one work week. The data is described in the table.*

<i>Time</i>	<i>Number of cars</i>
5:00-5:10	75
5:10-5:20	82
5:20-5:30	66
5:30-5:40	77
5:40-5:50	61
5:50-6:00	50

- What is the probability that a randomly selected car that went through the roundabout during the 5-6 PM hour entered it between 5:10 and 5:20 PM?*
- Find the probability of E =event that a car went through the roundabout between 5:00 and 5:30?*

Before listing the other properties, we need to review disjoint sets. Recall that two sets are disjoint when they have no elements in common. In the context of events, we use a new term to describe this.

Definition 192. Two or more events are **mutually exclusive** if they have no outcomes in common. In other words, E_1 and E_2 are mutually exclusive if

$$E_1 \cap E_2 = \emptyset$$

Example 193. E and $\bar{E}$ are mutually exclusive, since by definition the complement is everything not in E .

Problem 194. Come up with an example of 3 mutually exclusive events if the sample space you are considering is the set of odd numbers.

Problem 195. Suppose you have a normal deck of 52 cards. In this deck, there are four suits: diamonds, spades, hearts, and clubs.

- a) If a card is drawn at random, what is the probability that it is a spade?
- b) Consider the events that the card drawn is a $E_1 = \{\text{spade, diamond}\}$ or $E_2 = \{\text{heart}\}$. Are these events mutually exclusive? What are the probabilities of E_1 and E_2 ?
- c) What about the probability of the union $E_1 \cup E_2$?
- d) What is $\overline{E_1 \cup E_2}$? What is the probability of this complement?

Problem 196. In general, what is $P(E) + P(\bar{E})$? Your answer should be a number.

To summarize the last problem as a theorem...

Theorem 197. Given an event E , $P(\bar{E}) = \underline{\hspace{2cm}}$

Problem 198. A company needs to choose two people out of five to employ. Call them A, B, C, D, and E.

- a) What is $n(S)$?
- b) Let E be the event that C is one of the two people hired. What is $n(E)$?
- c) Find the probability that C is hired.
- d) Find the probability that C is not hired.

The next property is a really helpful one.

Theorem 199. (Inclusion-Exclusion Principle) If E and F are two events, then $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.

Problem 200. How does the above property compare to the counting formula? If you start with the counting formula, can you algebraically get to the Inclusion-Exclusion Principle?

Problem 201. Given A and B be events with $P(A) = 0.2$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$.

- a) Find the probability that A or B happens.
- b) Find the probability that A does not happen.
- c) Are A and B mutually exclusive? Why or why not?

Problem 202. As usual, politicians are worried about partisanship in the government. Suppose a new U.S. Senate committee was formed that had 3 of 8 members as Democrats. If there are 49 Democrats and 51 Republicans in the U.S. Senate and the committee of 8 was formed by selecting members randomly, what is the probability that the committee is composed of at most 3 Democrats?

You've probably heard people use the term "the odds of something are". Mathematically, the following tells you what that really means.

Definition 203. If E is an event, then

- The **odds for** E are $\frac{P(E)}{P(\bar{E})}$ read as $P(E)$ to $P(\bar{E})$.
- The **odds against** E are $\frac{P(\bar{E})}{P(E)}$ read as $P(\bar{E})$ to $P(E)$.

Problem 204. Suppose the probability that it will rain tomorrow is .3. What are the odds for rain?

Problem 205. Suppose the odds against the Chicago Cubs winning the pennant are 200 to 1. What is the probability that the Cubs win the pennant?

7.3 Expected Value

It makes sense that if someone wants to know the probability of an event that they would also want to know the expected value of it if a payoff is involved. For example, a gambler would want to know how much they should expect to win, on average.

Example 206. A fair coin is flipped twice. When two heads appear, you get \$2. When one head appears, you get \$1. If two tails appear, you lose \$4. What is the expected value?

Solution:

The sample space is $S = \{HH, HT, TH, TT\}$.
The probabilities of the three payoff events are:

$$P(HH) = \frac{1}{4} \quad P(HT \cup TH) = \frac{1}{2} \quad P(TT) = \frac{1}{4}$$

So, $\frac{1}{4}$ of the time, you would expect to win \$2.

Another $\frac{1}{4}$, you lose \$4.

And $\frac{1}{2}$ of the time, you expect to win \$1.

The expected value is $E = 2(\frac{1}{4}) + 1(\frac{1}{2}) - 4(\frac{1}{4}) = \frac{1}{2} + \frac{1}{2} - 1 = 0$.

If you played this game over and over, you should expect to neither make nor lose money.

When that happens, the game is called fair.

Definition 207. If each probability p_i has payoff m_i , then the **expected payoff**, E , is

$$E = m_1 \cdot p_1 + m_2 \cdot p_2 + \cdots m_n \cdot p_n$$

Definition 208. A game is called:

- *favorable* if $E > 0$
- *fair* if $E = 0$
- *unfavorable* if $E < 0$

Problem 209. You drive a \$20,000 car and currently make insurance payments of \$150 a month. If you damage your car in the first year of owning it, your insurance will cover the damages up to \$5,000.

- a) If the insurance company decides you have a 95% chance of not damaging your car in the first year, how much do they expect to profit off of your policy.
- b) What is the least amount of money the company could charge and still expect to make a profit off of your policy?

Problem 210. A student's grade in a certain Finite Mathematics class is based on weighted categories: 15% Participation, 20% Homework, 45% Tests, and 20% Final Exam. Suppose the student has scores of 95, 85, 88, and 91, respectively.

- a) What is the student's course grade?
- b) How is this calculation similar to the expected value formula? How is it different?

7.4 Conditional Probability

So far, when we have computed the probability of an event it was relative to the entire sample space: $P(E) = \frac{n(E)}{n(S)}$. When I ask for the probability $P(E)$ of an event E , this probability represents the likelihood that an experiment will produce an outcome in the set E relative to the sample space S . Note that we are no longer assuming that all simple events are equally likely.

Sometimes we want to compute the probability of an event E relative to another event F of the same sample space. For example, if we *know* that an outcome in F must happen, does that change the likelihood of E ? This type of probability is called a conditional probability because we are assuming the condition that F has occurred.

Definition 211. The **conditional probability** of an event E given event F , denoted $P(E|F)$, is the probability of E given that F has occurred.

Problem 212. A certain mathematics class contains 26 students. Of these, 14 are economics majors, 15 are freshmen, and 7 are neither. Suppose that a person is selected at random from the class. Let E =the event that an economic major is selected. Let F =the event that a freshman is selected.

- Draw a Venn diagram with two circles representing E and F . Include numbers for each part of the diagram to show how many students are in them.
- What is the probability that the person selected is both an economics major and a freshman? In other words, find $P(E \cap F) = \frac{n(E \cap F)}{n(S)}$
- What is the probability that the person selected is an economics major?

Suppose we know that the student selected is a freshman. Then the pool of possible students chosen is restricted.

- Instead of $n(S) = 26$, what is the size of our sample space now?
- What is $P(E|F)$, the probability that an economics major is picked if we know he or she is a freshman?

Problem 213. Consider the probability formula $P(E) = \frac{n(E)}{n(S)}$ and the restriction of possible events if event F is a given condition. How will this change the numerator and denominator? Using the last problem as a guideline, give a formula for the conditional probability $P(E|F)$:

$$P(E|F) =$$

Problem 214. What is $P(E|F)$ if E and F are mutually exclusive?

Problem 215. What is $P(E|F)$ if there is no chance that a simple event in F occurs (i.e. $P(F) = 0$)?

Problem 216. Twenty percent of the employees of a company have a master's degree. Of all its employees, 25% earn more than \$50,000 per year, and 15% of those with master's degrees earn more than \$50,000. Let E =event that an employee earns more than \$50,000 and F =event that an employee has a master's degree.

- a) What is $P(F)$?
- b) What is $P(E \cap F)$?
- c) What is the probability that an employee selected at random earns more than \$50,000 per year, given that he or she has a master's degree, or $P(E|F)$?

We can use the conditional probability formula to solve for more than just conditional probabilities. Notice that part of the formula includes the probability of an intersection. Then if you know the other two parts of the formula, the conditional probability and the denominator probability $P(F)$, you could solve for the intersection probability.

Problem 217. Assume that a marketing firm sends an equal number of surveys to people in Chicago and San Francisco. There is a 10% chance of a Chicagoan returning the survey and a 30% chance of a San Franciscan returning it.

- a) Make a tree diagram describing the events and probabilities of this problem.
- b) Find the probability that a randomly selected person who was sent the survey is from Chicago and returns it.

Problem 218. The table below contains probabilities of events A, B, C , and D . For example, .4 is the probability that both A and D occur, since it is the value in both the A column and D row.

	A	B
C	.3	.2
D	.4	.5

- a) Find $P(A)$.
- b) Find $P(A \cap C)$.
- c) Find $P(C|A)$.
- d) Find $P(A \cup C)$.

7.5 Independent Events

One of the most important concepts in probability is that of events being independent. This is when the fact that one event has occurred does not affect the probability of the other event occurring. So, even if you are given that event F has happened, the probability of event E is no different.

Definition 219. Let E and F be two events of a sample space S with $P(F) \geq 0$. The event E is **independent** of F if and only if

$$P(E|F) = P(E)$$

We know that $P(E \cap F) = P(E|F) \cdot P(F)$. If the events are independent, then $P(E|F) = P(E)$, and we get the following theorem.

Theorem 220. E and F are independent events if and only if $P(E \cap F) = P(E) \cdot P(F)$

Problem 221. Given $P(E) = 0.4, P(F) = 0.5, P(E \cup F) = 0.7$.

a) Find $P(E|F)$.

b) Are E and F independent?

Problem 222. Fill in the following blanks with the appropriate formulas for the probability of an intersection:

a) (Conditional Probability) $P(E \cap F) =$

b) (Inclusion-Exclusion Principle) $P(E \cap F) =$

c) If E and F are independent, then $P(E \cap F) =$

d) If E and F are mutually exclusive, then $P(E \cap F) =$

Problem 223. A Pew Research Center poll, conducted in 2018, found that 54% of college graduates either identified as Democrats or leaned Democratic. One-third of Americans have a college degree. Is this enough information to determine whether having a college education and being a Democrat are independent events?

Problem 224. A new running shoe helps runners fight arch pain for 80% of people. If four runners with arch pain wear the shoes, what is the probability that

a) all four runners' feet feel better?

Here is the start of the solution:

Let E_1 = event that the first runner's feet feel better.

Let E_2 = event 2nd runner's feet feel better.

Let E_3 = event 3rd runner's feet feel better.

Let E_4 = event 4th runner's feet feel better.

Then $P(E_1) = P(E_2) = P(E_3) = P(E_4) = 0.80$

We know that these events are independent because the fact that one runner feels better wouldn't affect the outcome of another runner.

Find $P(E_1 \cap E_2 \cap E_3 \cap E_4) = P(\text{all four better})$.

b) all four runners' feet don't feel better? (Hint: If E and F are independent then so are their complements)

7.6 Bayes' Theorem

The motivation for Bayes' Theorem is being able to switch the order of a conditional probability. Say that you know $P(E|F)$, or the probability of E given F , but you want to know $P(F|E)$, the probability of F given E .

For example, suppose we know the probability that someone makes more than \$50,000 a year given that the person has a college degree. What is the probability that a person has a college degree if we know that they make more than \$50,000 a year?

Bayes' Theorem gives us a way to answer that. We're going to develop the theorem through the following example.

Example 225. Suppose 86% of married women over 18 have at least 1 child, 31% of single (never married) women have at least 1 child, and 86% of previously married women (divorced or widowed) have at least 1 child. Also, 48% of women surveyed were currently married, 24% single, and 28% previously married. Find the probability of a woman having no children.

Let A_1 = currently married.

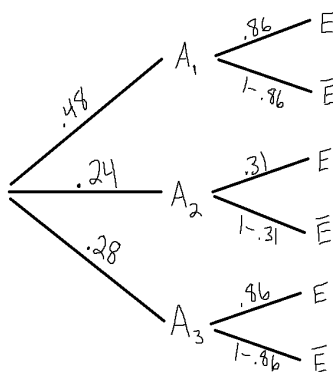
Let A_2 = single.

Let A_3 = previously married.

Let E = event that the woman had at least 1 child.

Then $\bar{E}$ = event of no children.

The tree diagram describes these probabilities.



Problem 226.

- In probability notation, what does .48 represent?
- In probability notation, what does .31 represent?
- What is $P(A_2 \cap E)$?

d) What is $P(E|A_3)$?

e) What is $P(\bar{E}) = P(A_1 \cap \bar{E}) + P(A_2 \cap \bar{E}) + P(A_3 \cap \bar{E})$? Hint: You are working through all the ways to get to $\bar{E}$ on the tree diagram. What numbers do you have to go through?

Now we're almost ready for Bayes' Theorem. Remember that the goal of the theorem is to find a way to "switch" the conditional probability.

We know that $P(E \cap A_i) = P(A_i)P(E|A_i)$ from the conditional probability formula.

Also, $P(A_i|E) = \frac{P(A_i \cap E)}{P(E)}$ from the conditional probability formula.

Since $P(E \cap A_i) = P(A_i \cap E)$, then we can substitute the first equation into the second to get:

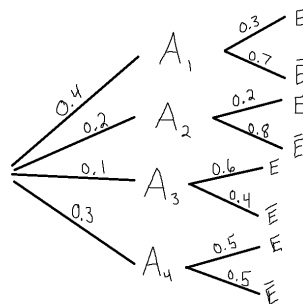
Theorem 227. Bayes' Theorem:

$$P(A_i|E) = \frac{P(A_i)P(E|A_i)}{P(E)}$$

Problem 228. Now find the probability that a woman who has one or more children is married. Hint: Remember that you already found $\bar{E}$ in a previous problem.

Try the following non-application example.

Problem 229. Given a tree diagram.



a) Find $P(E|A_3)$.

b) Find $P(E)$.

c) Find $P(A_3|E)$.

Note⁶

Chapter 8

Markov Chains

8.1 Markov Chains and Transition Matrices

In this section we are bringing the very useful mathematical tool of matrices back. It may seem like we are completely switching gears from the probability material, but there are connections. Specifically, we are going to look at experiments for which the probability of a certain outcome is the same for every time the experiment is repeated. Also, the possible outcomes never change.

A **Markov chain** is a model describing a sequence of possible events in which the probability of each event depends only on the state or condition attained in the previous event.

Note: The term ‘state’ here can represent many possible outcomes or conditions, depending on the context of the problem. In the following examples we will see that states may mean weather conditions or streaming services.

More formally, we have the terminology:

- The status of the person or object in the experiment is the “**state**” that it is in.
- The different trials are called “**events**” or “**observations**”.

Markov chains have applications in many areas, including business, psychology, sociology, and biology. Here are some situations that may be thought of as Markov Chains:

- Weather conditions change from day to day. For example, yesterday could have been sunny but today is rainy. There are two states here with movement between the states. Weather patterns can be tracked over time, so we could use that data to answer questions such as “What percentage of days this month will have rain?”

- Amazon Prime and Netflix are competing for viewers. Each week there is some shift of viewer loyalty from one platform to another. We could ask: What is the probability that a person would stop watching Netflix and start watching Amazon Prime?

Definition 230. Transition Probabilities: p_{ij} represents the probability of moving from state i to state j in one time step.

Definition 231. Transition Matrix: The transition matrix, P , is the $m \times m$ matrix displaying the transition probabilities.

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}$$

Example 232. Meijer recently opened a new store in town and initiated an extensive advertising campaign. After the campaign, a market research firm finds that there is a probability of 0.7 that a Meijer customer will return to Meijer and a .4 chance that a Target customer will switch to Meijer. Assume that the probability that a customer comes to a given store depends only on the previous store visited. Find the transition matrix representing this problem.

Solution:

If there is a .7 chance that a Meijer customer will return to Meijer, then there must be a $1 - .7 = .3$ chance that the customer will switch to Target.

If there is a .4 chance that a Target customer will switch to Meijer, then there must be a $1 - .4 = .6$ chance that a Target customer will stay with Target.

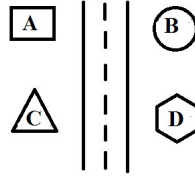
Let state 1=customer going to Meijer and state 2=customer going to Target.

Then the transition matrix is

$$P = \begin{bmatrix} .7 & .3 \\ .4 & .6 \end{bmatrix}$$

Example 233. There are 4 different car dealerships on a section of highway US 30. A customer has to choose which car dealership to purchase a new vehicle from. The dealerships are the states the customer can be in.

Let p_{ij} =probability that the customer moves from state i to state j . For example,



- p_{11} = probability that the customer stays at dealership A.
- p_{12} = probability that the customer leaves dealership A to shop at dealership B.
- p_{32} = probability that the customer leaves dealership C to shop at dealership B.

Problem 234. Give the transition matrix for the dealership example, using p_{ij} notation instead of numbers as the entries.

A common question arising in Markov chain models is “What is the long-term probability that the system will be in each state?”

The matrix containing these long-term probabilities is called the *steady-state vector* of the Markov chain. A vector is just a matrix with only 1 row or only 1 column.

Notationally, the vector is written

$$v = \begin{bmatrix} P(\text{end up in state 1}) & P(\text{end up in state 2}) & \cdots & P(\text{end up in state } m) \end{bmatrix}$$

Finding this vector, containing the probabilities of ending up in each state eventually, is our end goal. To get there, we must start at the beginning, with the *initial* probability vector.

$$v^{(0)} = \begin{bmatrix} P(\text{start in state 1}) & P(\text{start in state 2}) & \cdots & P(\text{start in state } m) \end{bmatrix}$$

Problem 235. *The Dealership Example:*

- What would the initial probability vector be if the customer previously bought a car at dealership A and we know he or she wants to look there first?
- What if the customer is equally likely to start at any of the four dealerships?

$v^{(1)}$ would be the probability distribution after the first observation. In other words, it tells us the probability of the second dealership that the customer will go to.

Generally, $v^{(n)} = \begin{bmatrix} P(\text{move to state 1}) & P(\text{move to 2}) & \cdots & P(\text{move to m}) \end{bmatrix}$ gives the probability of being in any given state after the n th trial.

How do we find $v^{(1)}$? Multiply $v^{(0)}P$.

How do we find $v^{(2)}$? Multiply $v^{(1)}P$.

How do we find $v^{(n)}$? Multiply $v^{(n-1)}P$.

Problem 236. *The Dealership Example:*

- a) *If the customer starts at dealership B, what is the probability vector after the 1st move (i.e. what is $v^{(1)}$)? Use the following transition matrix, which gives higher probabilities of staying at a dealership or going to a dealership on the same side of the street.*

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{2} & \frac{1}{8} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{8} & \frac{1}{2} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{8} & \frac{1}{2} \end{bmatrix}$$

- b) *Using your answer to part 1, what is the probability that the customer is at dealership D after one move?*

Problem 237. *In the 1960s, census studies revealed that 80% of the daughters of women with paid employment also had paid employment and 30% of the daughters of women without paid employment ended up with paid employment. Assume that this trend remains unchanged from one generation to the next.*

- a) *Determine the transition matrix if state 1 is that the woman has paid employment and state 2 is that she does not.*
- b) *Now suppose we know that in 1960 about 40% of women had paid employment. What is the probability distribution of the next generation?*

8.2 Regular Markov Chains

Remember our goal of finding a steady-state vector that describes the long-term probabilities of a Markov chain? Well, it ends up that this vector only exists for a certain type of Markov chain, called regular.

Definition 238. A Markov chain is said to be **regular** if, for some power of its transition matrix, P , all of the entries are positive.

This definition says that for SOME power of P , the entries are positive. This means no zero or negative numbers in the matrix. It could happen with P , P^2 , P^3 , or P^{100} . It just needs to be all positive for some power.

Problem 239. Decide whether the transition matrix is regular or not.

a) $\begin{bmatrix} .2 & .8 \\ 1 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 0 & .6 & .4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

What do the entries in these powers of P represent?

Theorem 240. If P is the transition matrix of a Markov chain, then the (i, j) th entry of P^n gives the probability of passing from state i to state j in n trials.

Example 241. Suppose $P^{12} = \begin{bmatrix} .5834 & .4166 \\ .5833 & .4167 \end{bmatrix}$. Then .5833 is the probability of moving from state 2 to state 1 in 12 trials.

Problem 242. Suppose $P^{10} = \begin{bmatrix} .58 & .42 \\ .3 & .7 \end{bmatrix}$. What is the probability of staying in state 2 after 10 trials?

For regular transition matrices, something special happens when you raise P to a high enough power. Eventually, the entries stop changing.

Example 243. Find the equilibrium matrix for $P = \begin{bmatrix} .7 & .3 \\ .4 & .6 \end{bmatrix}$ (This was the matrix for the first Markov problem.)

Solution:

$$P^2 = \begin{bmatrix} .61 & .39 \\ .52 & .48 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} .583 & .417 \\ .556 & .444 \end{bmatrix}$$

$$P^4 = \begin{bmatrix} .5749 & .4251 \\ .5668 & .4332 \end{bmatrix}$$

$$P^5 = \begin{bmatrix} .57247 & .42753 \\ .57004 & .42996 \end{bmatrix}$$

$$P^6 = \begin{bmatrix} .571741 & .4282596 \\ .571012 & .428988 \end{bmatrix}$$

We could find P^{1000} , but it would be basically the same as P^6 since the numbers are not changing much by that point.

The entries will never change after reaching equilibrium.

The long-term probability of moving to state 1 is .57 and the long-term probability of moving to state 2 is .43.

Theorem 244. Long-Term Behavior of a Regular Markov Chain

- The matrices P^n approach a fixed matrix F as n gets large.
- The rows of F are all identical and equal to a probability row vector v .
- v is the unique probability vector that satisfies $vP = v$.
- If $v^{(0)}$ is any initial distribution vector, then $v^{(n)} \approx v$ for large enough n .

The third item tells us that instead of having to find P^n for larger and larger values of n until the matrix stops changing, all we have to do to find the steady-state probability vector is solve the equation $vP = v$.

Problem 245. Let $P = \begin{bmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{2}{10} & \frac{8}{10} \end{bmatrix}$.

- Is this Markov chain regular? Why or why not?
- What size would the vector v need to be to be able to multiply vP ? Name the entries of v as $v_1, v_2, \dots$
- Set up the matrix equation $vP = v$. Multiply vP .
- Using your result from part c, you should be able to get 2 equations. Thinking back to chapter 1, what type of pair of lines are these?
- Using those two equations, plus the fact that $v_1 + v_2 = 1$, solve this system of equations.
- What is the probability of moving to state 1 in the long-run? State 2?

Problem 246. Given the transition matrix

$$P = \begin{bmatrix} .2 & .6 & .2 \\ .1 & .1 & .8 \\ .3 & .3 & .4 \end{bmatrix}$$

- a) Get a system of equations using the matrix equation $vP = v$.
- b) What do you think the best way to solve this system is?

Problem 247. Suppose the probability that it rains in a certain climate depends on the weather the day before. If it rains one day, the probability that it rains the next is $\frac{1}{4}$. If it is dry one day, the probability that it rains the next is $\frac{1}{2}$. Find the long-term probability distribution of whether it will rain or not the next day.

Problem 248. You are choosing which show to watch on Netflix. There are three shows that you currently enjoy: *Arrested Development* (A), *Breaking Bad* (B), and *The Crown* (C). If you watch A, then the next day you watch B. However, if you watch either B or C, then the next day you are three times as likely to watch A as the other show.

- a) Find the transition matrix for this Markov chain.
- b) In the long run, what is the probability of you watching each show? Get to the point where you have an augmented matrix set up.
- c) The row reduced augmented matrix is below. Finish the problem to find the probability vector which tells you the probability she watches each of the three shows in the long run.

$$\left[\begin{array}{ccc|c} 1 & -\frac{3}{4} & -\frac{3}{4} & 0 \\ 0 & 1 & -4 & 0 \\ 0 & 0 & 1 & \frac{4}{35} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

- d) What is the probability that you watch *Arrested Development* in the long run?

8.3 Absorbing Markov Chains

What happens if a transition matrix is not regular?

That means that no matter what power n you take the transition matrix to, there will still be negative or zero entries in the matrix. For a subset of these Markov chains, called **absorbing Markov chains**, we can still find out a lot about what will happen in the long run.

A state is called **absorbing** if once you have moved into the state, you can never leave it. It “absorbs” the object, or whatever is involved in the experiment.

Problem 249. Given a transition matrix $P = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ 0 & 1 \end{bmatrix}$, which state is the absorbing state and why?

Definition 250. In a Markov chain let p_{ii} denote the probability of remaining in state E_i .

- If $p_{ii} = 1$, then E_i is called an **absorbing** state.
- A Markov chain is an **absorbing Markov chain** if and only if it contains at least one absorbing state and it is possible to go from any nonabsorbing state to an absorbing state in one or more trials.

Problem 251. Find the absorbing state and determine whether the chain is an absorbing Markov chain. The labels E_i on the left tell you which state you are coming from. The labels E_i on top tell you which state you are going to.

a)

$$\begin{array}{c} E_1 \quad E_2 \quad E_3 \quad E_4 \\ \begin{array}{l} E_1 \\ E_2 \\ E_3 \\ E_4 \end{array} \left[\begin{array}{cccc} 0 & .5 & .5 & 0 \\ .2 & .3 & .1 & .4 \\ 0 & 0 & 1 & 0 \\ .1 & 0 & .9 & 0 \end{array} \right] \end{array}$$

b)

$$\begin{array}{c} E_1 \quad E_2 \quad E_3 \\ \begin{array}{l} E_1 \\ E_2 \\ E_3 \end{array} \left[\begin{array}{ccc} 1 & 0 & 0 \\ .3 & .4 & .3 \\ .2 & .5 & .3 \end{array} \right] \end{array}$$

c)

$$\begin{array}{c} E_1 \quad E_2 \quad E_3 \\ \begin{array}{c} E_1 \\ E_2 \\ E_3 \end{array} \left[\begin{array}{ccc} 1 & 0 & 0 \\ .3 & .4 & .3 \\ 0 & 0 & 1 \end{array} \right] \end{array}$$

The big question for absorbing Markov chains is “How many trials does it take to get absorbed?”

The first step to figuring this out is to divide the transition matrix into absorbing states and nonabsorbing states. We want to rearrange the the rows and columns so that the absorbing states are in the top left of the matrix.

Example 252. *Subdivide the transition matrix into absorbing states and nonabsorbing states.*

$$T = \begin{array}{c} E_1 \quad E_2 \quad E_3 \quad E_4 \quad E_5 \\ \begin{array}{c} E_1 \\ E_2 \\ E_3 \\ E_4 \\ E_5 \end{array} \left[\begin{array}{ccccc} 0 & .5 & 0 & .25 & .25 \\ 0 & 1 & 0 & 0 & 0 \\ .7 & .3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ .2 & .2 & .2 & .2 & .2 \end{array} \right] \end{array}$$

Solution:

E_2 and E_4 are the absorbing states, so I want to move those to the first two rows/columns.

$$\begin{array}{c} E_2 \quad E_4 \quad E_1 \quad E_3 \quad E_5 \\ \begin{array}{c} E_2 \\ E_4 \\ E_1 \\ E_3 \\ E_5 \end{array} \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline .5 & .25 & 0 & 0 & .25 \\ .3 & 0 & .7 & 0 & 0 \\ .2 & .2 & .2 & .2 & .2 \end{array} \right] \end{array}$$

Each p_{ij} entry in T matches the E_i row and E_j column entry in the rearranged matrix.

Notice that the top left quadrant looks like the identity matrix, I_2 . The top right is the zero matrix.

Bottom left is a 3×2 matrix. Let's call it S .

$$S = \begin{bmatrix} .5 & .25 \\ .3 & 0 \\ .2 & .2 \end{bmatrix}$$

Bottom right is a 3×3 matrix. Let's call it Q .

$$Q = \begin{bmatrix} 0 & 0 & .25 \\ .7 & 0 & 0 \\ .2 & .2 & .2 \end{bmatrix}$$

When a matrix has been subdivided into the absorbing states (listed first) and the nonabsorbing states, it will always have the same form.

Let r = number of absorbing states.

Let q = number of nonabsorbing states.

$$P = \left[\begin{array}{c|c} I_{r \times r} & 0_{q \times r} \\ \hline S_{q \times r} & Q_{q \times q} \end{array} \right]$$

Problem 253. Subdivide the transition matrix $P = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ of the

Markov chain.

Once you know what S and Q are, you can find what's called the **fundamental matrix**, T . This matrix will tell us the expected number of times the process is in each nonabsorbing state before getting absorbed into one of the absorbing states.

Here is how to find the fundamental matrix, T :

$$T = [I - Q]^{-1}$$

Problem 254. Using the previous example's $Q = \begin{bmatrix} 0 & 0 & .25 \\ .7 & 0 & 0 \\ .2 & .2 & .2 \end{bmatrix}$:

a) Find $I - Q$.

b) Set up the augmented matrix that you would use to find the inverse of $I - Q$.

The final result for this inverse problem should be

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1.119 & 0.07 & 0.35 \\ 0 & 1 & 0 & 0.783 & 1.049 & 0.245 \\ 0 & 0 & 1 & 0.476 & 0.28 & 1.399 \end{array} \right]$$

Then the fundamental matrix is

$$T = [I - Q]^{-1} = \begin{bmatrix} 1.119 & 0.07 & 0.35 \\ 0.783 & 1.049 & 0.245 \\ 0.476 & 0.28 & 1.399 \end{bmatrix}$$

The first row corresponded to state E_1 in the subdivided matrix, that row gives the expected number of trials before absorption if the process starts in E_1 .

$$1.119 + 0.07 + 0.35 = 1.539$$

Since you can't have a fraction of a trial, round up to 2. If the process starts in E_1 , absorption is expected to happen after 2 transitions.

Similarly, look at the second row for state E_3 .

$$0.783 + 1.049 + 0.245 = 2.077$$

If the process starts in E_3 , absorption is expected to happen after 3 rounds (rounding up).

Look at the third row, which corresponds to state E_5 .

$$0.476 + 0.28 + 1.399 = 2.155$$

If the process starts in E_5 , absorption is expected to happen after 3 rounds.

This matrix does not tell us which absorbing state, E_2 or E_4 , ends up being the one that absorbs. The process can only be absorbed into one state. How do we know which one it is?

The best answer to the second question can be found in the matrix product, $T \cdot S$. The (i, j) entry of $T \cdot S$ gives the probability of the state corresponding to column j being the absorbing state given that the process starts in nonabsorbing state i .

Example 255. Previous example continued: Find the probability of going from starting in state E_1 to being absorbed into E_2 ?

Solution:

$$T \cdot S = \begin{bmatrix} 1.119 & 0.07 & 0.35 \\ 0.783 & 1.049 & 0.245 \\ 0.476 & 0.28 & 1.399 \end{bmatrix} \begin{bmatrix} .5 & .25 \\ .3 & 0 \\ .2 & .2 \end{bmatrix} = \begin{bmatrix} .686 & .3675 \\ .7572 & .2448 \\ .6018 & .3988 \end{bmatrix}$$

Then the probability is .686.

Problem 256. Given the fundamental matrix and the matrix S find the probability that if the process starts in state E_4 it will be absorbed into state E_1 .

$$T = \begin{matrix} & \begin{matrix} E_2 \\ E_4 \end{matrix} \end{matrix} \begin{bmatrix} 1 & .75 \\ 0 & 1 \end{bmatrix}$$

$$S = \begin{matrix} & \begin{matrix} E_1 & E_3 \end{matrix} \end{matrix} \begin{bmatrix} .25 & 0 \\ .1 & .9 \end{bmatrix}$$

Problem 257. Given the following transition matrix, what is the expected number of trials before absorption if the initial state is E_1 ? E_2 ? What is the probability that E_3 is the absorbing state for each of these initial states?

$$\begin{matrix} & \begin{matrix} E_1 & E_2 & E_3 \end{matrix} \\ \begin{matrix} E_1 \\ E_2 \\ E_3 \end{matrix} & \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Problem 258. You need to write a paper for a class. In this class, students are given the opportunity to complete multiple drafts of a paper in order to improve their grade. If at any point a student receives an A or B on a draft, they can no longer receive a worse grade. Historically, if a student receives a C or D on a draft, there is a 70% chance they receive a B or better on the next draft, and a 30% chance they stay at the same level. If a student receives a failing grade (F) on a draft, there is a 10% chance they are able to get a B or better on the next draft, a 60% chance they receive a C/D on the next draft, and a 30% chance they still receive an F.

a) Find the transition matrix for this Markov Chain. Which state(s) are the absorbing states?

- b) Subdivide the transition matrix.*
- c) How many trials would you expect the process to take before absorption?*
- d) What is the probability that if a person starts out with a failing grade, that he or she will eventually get an A or B?*

Note⁷

Chapter 9

Two-Person Games

9.1 Strictly Determined Games

Now you are going to learn about games! We are going to look specifically at games with two players or teams. These problems don't necessarily have to be "play" games. We could be studying the strategies of two sides of a war, or two politicians' campaigns.

Definition 259. *Any conflict or competition between two people is called a two-person game.*

Goal: How to choose the best strategy in a two-person game?

Problem 260. *Stores R and C decide to have a sale. Both could give 10% off or 20% off. Suppose they start with each having 50% of the customers. Their past history is:*

- *If both choose 10% off, then R gets 60% of the customers.*
 - *If R chooses 10% and C chooses 20%, then R gets 35% of the customers.*
 - *If R chooses 20% and C chooses 10%, then R gets 80% of the customers.*
 - *If both choose 20%, then R gets 50% of the customers.*
- a) *Represent R's gains and/or losses with a matrix, where each row represent store R's choices and each column represents store C's choices.*
- b) *Which sale option do you think R should choose? What about C?*

We call the matrix describing the choices for each player, and the outcomes for those choices, a **payoff matrix**. As in the above example, each row represents the options of one player and each column represents the

options for the other player. Then an entry in the matrix is the outcome, in terms of the row player's gains or losses, if they choose the options corresponding to the row and column containing that entry.

Definition 261. A *zero-sum* game is when the gains of one player equal the losses of the second player.

There is a type of zero-sum, two-person game for which we can immediately know the best strategies. This type is called **strictly-determined**.

Definition 262. A game is *strictly-determined* if and only if there is an entry in the payoff matrix that is the smallest in its row and the largest in its column.

A few important terms:

- This entry (the smallest in the row/largest in the column) is called the **saddle point**.
- The saddle point gives the **value** of the game. The value represents the expected gain/loss of the row player if both players choose their best strategy.
- The best strategy for each player is their **pure optimal strategy**. This strategy tells the player what row or column they should always choose.
- If the value is 0, the game is called **fair**.

Problem 263. *Previous problem continued:*

- a) *Is this a zero-sum game?*
- b) *This game is strictly determined. What is the saddle point, or value, of the game?*
- c) *The row that the saddle-point is in is the row that store R should choose. Does that match your previous answer? What about the column for C?*

Problem 264. *Is the game with the given payoff matrix strictly determined? If so, what strategies should R and C choose?*

$$a) \begin{bmatrix} 3 & 0 & -2 & -1 \\ 2 & -3 & 0 & -2 \\ 4 & 2 & 1 & 0 \end{bmatrix}$$

$$b) \begin{bmatrix} 1 & 2 & 1 \\ 1 & 5 & 1 \\ 0 & -7 & -1 \end{bmatrix}$$

Problem 265. Two players show heads or tails of a coin at the same time. The payoff matrix, where the entries represent dollars, is given below.

$$\begin{array}{c}
 \text{C (columns)} \\
 \\
 \begin{array}{cc}
 & H & T \\
 \begin{array}{c} R \text{ (rows)} \\ H \\ T \end{array} & \begin{bmatrix} 5 & -8 \\ 3 & 1 \end{bmatrix}
 \end{array}
 \end{array}$$

- Interpret the matrix. Describe what each number represents.
- What are their pure optimal strategies (the strategies they should choose every time)?
- What is the value of the game and what does it mean?

Problem 266. Two candidates for political office must decide to be for, against, or neutral on a certain topic. Pollsters have determined that if both candidates R and C come out in favor, then R gains 8000 votes. If R is in favor and C is against, then R loses 1000 votes. If R is in favor and C is neutral, then R gains 1000 votes. If R is against and C is in favor, then R loses 7000 votes. If both are against, then R gains 4000. If R is against and C is neutral, then R loses 2000. If R is neutral, he will gain 3000 votes if C is for or against and will gain 2000 if C is neutral. Determine the payoff matrix and the pure optimal strategies for each candidate.

9.2 Mixed Strategies

Not every game has a saddle-point and is strictly-determined. Sometimes there is no pure optimal strategy for a player, meaning there is no single strategy that they should choose every time the game is played. Maybe they should choose strategy 1 part of the time and strategy 2 part of the time.

Definition 267. *Mixed strategies* are where a player chooses their strategies with a particular probability.

Remember that a game is not always played just once. These probabilities give a strategy for the long-term.

Suppose we know the probability that a player wants to choose a certain strategy.

In a game with two choices, we could represent the strategy for R by $P = \begin{bmatrix} p_1 & p_2 \end{bmatrix}$.

- p_1 is the probability that R should choose row 1 (strategy 1).

- p_2 is the probability that R should choose row 2 (strategy 2).

In a game with two choices, we could represent the strategy for C by

$$Q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}.$$

- q_1 is the probability that C should choose column 1 (strategy 1).
- q_2 is the probability that C should choose column 2 (strategy 2).

The generic payoff matrix for R is $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.

Problem 268. *In what order should we multiply these matrices so that the probabilities for the rows of A are multiplied through the rows, and the probabilities for the columns are multiplied through those columns? What size matrix would you end up with in the end? That matrix represents the expected payoff for R.*

Problem 269. *Given $A = \begin{bmatrix} 2 & -2 \\ 0 & 1 \end{bmatrix}$, $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, $Q = \begin{bmatrix} \frac{4}{5} \\ \frac{1}{5} \end{bmatrix}$. Find the expected payoff, E , for player R.*

Problem 270. *Two players simultaneously show a fist (rock), a flat hand (paper) or two fingers (scissors).*

Rock beats scissors.

Scissors beat paper.

Paper beats rock.

Suppose the loser pays a penny to the winner.

a) *Find the payoff matrix using order rock, paper, and then scissors.*

b) *Suppose R's mixed strategy is $P = \begin{bmatrix} 0.4 & 0.2 & 0.4 \end{bmatrix}$. Should C choose*

$$Q_1 = \begin{bmatrix} 0.1 \\ 0.5 \\ 0.4 \end{bmatrix} \text{ or } Q_2 = \begin{bmatrix} 0.9 \\ 0 \\ 0.1 \end{bmatrix} ?$$

9.3 Finding Optimal Mixed Strategies

We saw in the last section that not all strategies will be pure optimal strategies, meaning the players won't always choose the same strategy every time

because the game is not strictly-determined. Sometimes their strategies will be mixed. Now we will figure out how to determine the probabilities in the mixed strategies, as in the probability that R should choose row 1 or row 2.

Even if a game is not strictly determined, we can still find an optimal strategy for each player.

Consider a game where each player has only two options.

Goal: Find the optimal mixed strategies $P = \begin{bmatrix} p & 1-p \end{bmatrix}$ for player R and $Q = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for player C.

We want to find the best expected value E for the two players. Let E_R be the expected value for R and E_C the expected value for C. Before, we found the expected value of the game by finding $E = PAQ$. Now we want to be more specific and only multiply certain parts of those matrices to get E_R and E_C .

Problem 271. Given the payoff matrix $A = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$

- Is this game strictly-determined?
- Let $P = \begin{bmatrix} p & 1-p \end{bmatrix}$ represent the mixed optimal strategy for R. Multiply PA to get a matrix for E_R . How many numbers/entries does this matrix have? We can only have one expected value for player R, so what do you think you should do with those entries?
- Let $Q = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ represent the mixed optimal strategy for C. Multiply AQ to get a matrix for E_C . Follow similar steps as you did for E_R .
- Once you have solved for P and Q , find $E = PAQ$ to get the overall expected value of the game. Is the game fair?

Problem 272. Find the optimal strategy for each player, and determine the expected payoff of the game given the payoff matrix $\begin{bmatrix} 6 & 0 \\ -2 & 3 \end{bmatrix}$.

Problem 273. Taylor Swift and Billie Eilish are releasing new albums on the same date. In preparation for this, they each have to decide whether they want to advertise on radio stations or online music streaming services. If both advertise on the radio, Taylor will sell 8,000 more records than Billie. If Billie instead advertises online, Taylor will sell 2,000 records less than

Billie. If both advertise online, Taylor will sell 3,000 more albums than Billie. Finally, if Billie buys radio ads and Taylor buys online ads, Taylor will sell 1,000 more. Find the optimal strategies for Taylor and Billie. Who is expected to sell more albums, and how many more?

Notes to the Instructor

¹In class, each group makes up three systems and then trades their systems with another group. Then each group checks to make sure that all three types are included. Finally, I have a couple of groups put the systems and solutions on the board for discussion.

²Students are given time to conjecture about this and then a couple of them write their matrices on the board. I then step in to discuss how an augmented matrix should look, building on their examples.

³I do not teach the 2×2 inverse formula in this class to avoid making the students memorize more formulas.

⁴After working on this for a short time, students write answers on the board. Then the instructor builds off those answers to describe the initial simplex tableau.

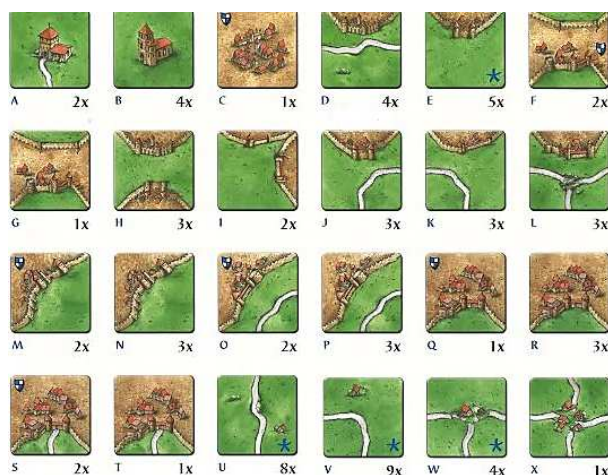
⁵I hand out dice in class and let the students actually roll them.

⁶At the end of this section I devote a couple of days to playing the board game *Carcassonne* and completing the associated problems seen in Appendix A.

⁷At the end of this chapter I devote a day to playing the board game *Chutes and Ladders* and completing the associated problems seen in Appendix B.

Appendix A

CARCASSONNE



These are all the tiles included in the game. There are a total of 72 tiles.

Goal: Get the most points. The game ends when you run out of tiles.

How to get points: You will place a person, or meeple, on each of the following to claim it. If you finishing building the road/city/monastery, then you get your person back and you get the points. If you don't finish building it, you get some or all of the points when the game ends.

- Roads- each piece of road counts as one point. A road is completed if each end has a piece that breaks up the road segment somehow.
- Cities- For a completed city, each piece is 2 points and each banner is 2 points. If uncompleted, each is 1 point.
- Monasteries- A monastery is complete when it is surrounded by tiles, which means there are 9 tiles including the monastery itself. Each tile counts as 1 point.

You cannot occupy a road, city, or monastery that someone else has claimed. If two separate roads or cities are joined, and both are claimed, then both players get the points.

After playing the game, answer the following:

- a) The monastery is a key type of tile because you can get up to nine points from this one tile. What is the probability of choosing a monastery if no tiles have been chosen yet?
- b) What is the probability of choosing a monastery if 20 tiles have been picked and two of them were monasteries?
- c) At the point in the game described in the last question, what are the odds of choosing a tile that is not a monastery?
- d) Suppose all of the banner pieces have been picked as well as two tiles with just a straight road (type U) and four tiles with just a curvy road (type V). What is the probability of choosing a tile that has a road or a monastery? What about a road and a monastery?
- e) At a given point in the game, assume that the following tiles are left to randomly choose from:
 - 3 monasteries (worth 5 points each because any of the three could be placed next to 4 other tiles)
 - 21 tiles with a city but no banner (worth 2 points each)
 - 4 tiles with a city and a banner (worth 4 points each)
 - 6 tiles with only roads (worth 1 point each)

Using the points described above, what is the expected value of your next tile?

- f) At the beginning of the game, how many ways are there for the first four tiles that are picked to be a monastery, then a tile with some portion of a city, and finally a tile with only a road (no monastery or city)? What is the probability that this happens?
- g) (Bayes' Theorem) Consider the following five mutually exclusive types of tiles: monasteries (M), cities with a banner (B), cities with a road but no banner (A), cities with no road or banner (C), and tiles with only roads (R).
 - (a) Start a tree diagram with these events/tiles as the first branches. Include the probability of each event.
 - (b) Your opponent, MacDonald, is deciding whether to place his meeple on a field (the grass) or the other important feature on each tile. MacDonald never chooses a field over a monastery. Banner pieces lead to higher points, so he only chooses to place his meeple on the field 3 times out of 10 for type B . Sixty percent of the time he

chooses the field for type A and half the time on C . Finally, three-fourths of the time he chooses fields over simple roads. Finish the tree diagram based on this information.

- (c) If you start a new game with player MacDonald where he gets to play a tile first, what is the probability that a road (R) was drawn if he placed a farmer on the tile?
- (d) Are the events of MacDonald drawing a road tile (R) and choosing a field independent?

Appendix B

CHUTES AND LADDERS

- Chutes and Ladders is a board game where players spin a pointer to determine how they will advance
 - The board consists of 100 numbered squares. The objective is to land on square 100.
 - The spin of the pointer determines how many squares the player will advance at his turn with equal probability of advancing from 1 to 6 squares.
 - Chutes have villains on them and will move a player back. Ladders have super heroes and will move a player forward.
- a) Why can this game be described using a Markov chain?
- b) What do you think the states are for this example?
- c) What size do you think the transition matrix should be?
- d) What would the first number in the transition matrix represent? What is that number?
- e) If you add up each number in row 1 (ignore the chutes and ladders), what would the sum be? What happens if we include the chutes and ladders?
- f) Could you have a negative number in the transition matrix?
- g) What is the initial probability vector for this game?
- h) Do you think this is a regular Markov chain or an absorbing Markov chain? Defend your answer. Can you think of a board game that would represent the other type of Markov chain?
- i) After 1000 moves, the probability vector for Chutes and Ladders goes to $v^{(1000)} = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix}$. What does this tell us about how long it takes to win the game?